Banking Fragility and the Capitalization Dilemma*

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April 2026

Abstract

Bank capital plays a dual role: abundant capital lets banks lever up and expand credit, amplifying productivity shocks, while scarce capital leaves the banking system endogenously fragile and crisis-prone. We embed both channels in a single general-equilibrium model and analyze it globally. Their interaction produces a capitalization dilemma: the effect of bank capital on macroeconomic stability is non-monotonic, a feature invisible to single-channel models. A social planner who internalises the fragility externality holds more capital, sharply cutting the frequency and cost of financial crises while leaving the economy more sensitive to productivity shocks. We distill the planner's solution into a simple time-consistent Taylor-rule-type leverage formula that tightens in credit-driven booms but accommodates booms driven by strong fundamentals. It responds to the financial and business cycles separately, which the credit-to-GDP gap conflates.

Keywords: Banking fragility, bank capitalization, macroprudential policy, occasionally binding constraints, global nonlinear solution.

JEL codes: E32, E44, G21, G28

* Any views expressed here are solely those of the authors and do not represent those of the Bank of England or any of its committees.
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1 Introduction

Bank capital requirements lie at the heart of macroeconomic and financial stability debates. Systemic buffers such as the Basel III countercyclical capital buffer are already part of the macroprudential toolkit, but their optimal level, timing and state-contingency remain contested. The policy puzzle is simple. Thinly capitalised banks are fragile: they are more exposed to failures, runs and funding stress when confidence falls. But well-capitalised banks can also expand credit more aggressively in good times, increasing the economy's exposure when adverse real shocks arrive. The banking turmoil of early 2023 (Gruenberg, Martin J., 2024; Basel Committee on Banking Supervision, 2023) brought both forces into view. It showed that capital matters for resilience, but also that the design and communication of buffers remain central policy questions. How high should bank capital requirements be, and when should they tighten or relax, if the same buffer can reduce systemic fragility while changing the transmission of normal downturns?

This paper fills that gap by studying both effects in one general-equilibrium model. The contributions of this paper are threefold. First, we show that combining a *fragility channel* (high leverage heightens run risk) with a *credit-exposure channel* (strong balance sheets amplify real shocks) gives rise to a fundamental *capitalization dilemma*. In our model, any fixed capital requirement involves a trade-off: higher bank capital significantly improves resilience to financial crises (reducing the frequency and severity of crises) but simultaneously increases banks' exposure to real-sector downturns (as they build larger balance sheets and intermediate more in good times). In other words, the relationship between bank capital and macroeconomic stability is *non-monotonic*—a novel insight that would be missed in models focusing on only one channel.

Second, we quantify the gap between privately optimal and socially optimal capital levels. A social planner internalising the fragility externality would choose meaningfully higher bank capital buffers than the private competitive equilibrium. This extra capital leads to lower crisis incidence and reduces output losses during financial crises, yet it leaves the economy more sensitive to ordinary productivity-driven recessions. Interestingly, the planner's intervention has little effect on purely real (non-financial) downturns, indicating that the gains from regulation come almost entirely from mitigating financial crises.

The third contribution is policy implementation. We derive a practical state-contingent leverage rule that approximates the planner's allocation. The rule tightens when balance-sheet expansion raises fragility and relaxes when strong fundamentals reduce the marginal externality. This separates the financial cycle from the business cycle, a distinction that simpler fixed capital ratios or credit-to-GDP gap triggers can obscure.

Underpinning these contributions is a quantitative model that combines ingredients from two strands of literature. We incorporate an endogenous *fragility (run) risk* that increases when bank equity is scarce, as in the banking-crisis models of Gertler and Kiyotaki (2010) and others. At the same time, banks in our model face an occasionally binding leverage constraint, so that high net worth translates into a lending boom and stronger transmission of real shocks (as in Brunnermeier and Sannikov (2014) among others). By integrating these two channels within one framework, we capture interactions that would be missed if each mechanism were studied in isolation. This integration is crucial for understanding why bank capital requirements entail a trade-off: ample capital acts as insurance against funding crises, but also supports larger balance sheets and greater macroeconomic sensitivity.

These mechanisms are not merely theoretical. Drawing on the long-run cross-country panel of Jordà et al. (2017), Section 2 documents three patterns that motivate the model's two channels: thinly capitalised banking systems are more prone to distress; better-capitalised systems tend to expand credit more rapidly; and economies hit by adverse shocks after a credit boom suffer deeper output losses. Each fact maps to one link in the dual role of bank capital—a buffer in rare crises, an amplifier of ordinary shocks—which the remainder of the paper formalises, quantifies, and translates into a macroprudential rule.

Related literature. This paper connects several strands of work on financial intermediation, banking fragility, and macroprudential policy.

It first builds on models of *banking panics and fragility*. Self-fulfilling depositor runs date to Diamond and Dybvig (1983), with the run probability microfounded as a global-games equilibrium by Goldstein and Pauzner (2005). Quantitative macroeconomic models embed such panics in general equilibrium: Gertler et al. (2020) generate endogenous runs that amplify downturns, Boissay et al. (2016) derive banking crises endogenously from credit booms, and He and Xiong (2012) study rollover risk in intermediary funding. We adopt a tractable reduced-form hazard $\mathcal{H}(N/L)$—disciplined by the Jordà et al. (2017) crisis database—that captures this literature's central comparative static (fragility rises as capitalization falls) while remaining embeddable in a globally-solved nonlinear model. Our aim is not a new theory of runs but an analysis of how run risk *interacts* with the real-amplification role of bank capital.

Second, it relates to the literature on *intermediary balance sheets and the financial accelerator*, in which net-worth fluctuations amplify and propagate shocks (Kiyotaki and Moore, 1997; Bernanke et al., 1999; Gertler and Kiyotaki, 2010; Brunnermeier and Sannikov, 2014;

He and Krishnamurthy, 2013, 2019), with leverage moving procyclically (Adrian and Shin, 2010; Gertler and Karadi, 2011) and intermediary capital priced as a state variable (Rampini and Viswanathan, 2019). These models stress that strong balance sheets stabilize the financial sector. Our *credit-exposure channel* shares the mechanism but draws the opposite-signed implication for real shocks: the same net worth that buffers fragility raises the scale of intermediation, so better-capitalized banks transmit *larger* productivity shocks. The novelty is that both forces operate through one state variable—bank equity—giving capital a *non-monotonic*, dual role that neither literature isolates on its own.

Third, it contributes to *quantitative analyses of bank capital requirements*. Van den Heuvel (2008) measures the welfare cost of requirements through forgone liquidity provision; Begenau (2020) shows requirements can raise welfare by reshaping banks' funding and risk choices; Corbae and D'Erasmo (2021) study capital buffers with banking-industry dynamics; and Elenev et al. (2021) embed financially-constrained intermediaries in a general-equilibrium model to evaluate macroprudential policy. These papers share our focus on welfare-relevant capital regulation and the trade-off between resilience and intermediation. Our contribution is to show that the trade-off is non-monotonic once crisis fragility and real-shock exposure operate through the same state variable: requirements that lower crisis risk can simultaneously raise exposure to productivity shocks. We also characterize both the constrained-efficient allocation and an implementable state-contingent leverage rule in a globally solved model that handles the occasionally binding leverage constraint exactly.

Fourth, it speaks to the *macroprudential policy and externalities* literature. A large body of work traces inefficient borrowing to *pecuniary* externalities operating through asset prices and collateral values (Lorenzoni, 2008; Bianchi, 2011; Mendoza, 2010), with optimal time-consistent policy characterized by Bianchi and Mendoza (2018) and the role of nominal rigidities by Farhi and Werning (2016). Our externality is different in kind: it is *compositional*, not pecuniary. Each atomistic bank takes the aggregate failure hazard as given, yet its leverage and equity choices move the sector-wide capital ratio N/L that determines that hazard; no relative price mediates the spillover. The planner corrects it with a Pigouvian wedge on leverage, which we map into a state-contingent leverage rule on bank equity. The closest policy antecedent is the optimized macroprudential debt tax of Bianchi and Mendoza (2018); our instrument instead maps directly to Basel-style capital requirements and is estimated over the nonlinear region around the leverage kink, where the credit-growth triggers of frameworks such as the Basel III countercyclical buffer (Basel Committee on Banking Supervision, 2010) conflate the financial and business cycles.

The remainder of the paper is organized as follows: Section 2 motivates the model with stylised facts, Section 3 presents the model, Section 4 discusses quantitative results, Section 5 analyses the constrained-efficient allocation and optimal policy rule, and Section 6 concludes.

2 Empirical facts

This section documents three reduced-form facts that motivate the model's two-channel mechanism. Fact 1 disciplines the fragility channel by linking lower system-wide capitalization to a higher probability of banking distress. Facts 2 and 3 discipline the credit-exposure channel: stronger capitalization predicts faster subsequent credit growth, and negative productivity shocks are more contractionary when they arrive after a credit boom. The empirical exercises use annual macro-financial panel data from the historical database of Jordà et al. (2017). For the productivity-shock interaction exercise, we merge the JST panel with Penn World Table TFP growth to identify negative productivity shocks. All three exercises are estimated at the country-year level with panel specifications that include standard macro controls and fixed effects.

2.1 Bank distress probability and lagged capital ratios

Fact 1: Less-capitalized banking systems have a significantly higher probability of distress. We estimate this relationship with a pooled logit of a pre-crisis indicator on the lagged aggregate capital ratio, using 2,259 country-year observations (154 pre-crisis episodes) from the JST panel. The estimated slope on lagged capital is strongly negative ($\beta = -6.86$ at annual frequency), implying that weaker capitalization is associated with materially higher distress probabilities. Table 1 reports the evidence. At the sample mean capital ratio (15.5%), the fitted pre-crisis probability is 5.8%, and the marginal effect is economically large: a one percentage point increase in the aggregate capital ratio is associated with roughly a 0.38 percentage point reduction in pre-crisis risk. This evidence motivates the hazard block in the model, where fragility decreases with capitalization.

2.2 Healthier balance sheets and subsequent credit growth

Fact 2: Better-capitalized banking systems subsequently experience faster credit growth. We estimate this relationship using country-panel local projections and find positive effects over subsequent years, with the strongest responses in cumulative specifications. Figure 1 shows that after an increase in aggregate bank capital ratios, credit-to-GDP tends

Table 1: Bank distress risk and lagged aggregate capital ratio: baseline logit summary

n_{obs}	n_{pre}	α (annual)	β (annual)	$\hat{p}$ at mean capital	Marginal effect at mean
2259	154	-1.718	-6.860	0.058	-0.376
		(0.159)	(1.207)		

Notes: Pooled logit of a pre-crisis indicator on the lagged aggregate bank capital ratio (JST panel). Standard errors in parentheses; both coefficients are significant at the 1% level ($p < 10^{-7}$), with 95% confidence intervals $\alpha \in [-2.03, -1.41]$ and $\beta \in [-9.23, -4.50]$. $\hat{p}$ denotes the fitted crisis probability at the sample-mean capital ratio (15.5%); the marginal effect is $\partial\hat{p}/\partial(N/L)$ at the mean (per unit of the ratio; equivalently about -0.038 per 10 percentage points).

to rise in subsequent years. This pattern is consistent with the model's credit-exposure channel: stronger bank balance sheets support more credit creation, which can amplify later fluctuations.

$$\Delta_h \text{CreditBoom}_{i,t+h} = \alpha_i^{(h)} + \gamma_t^{(h)} + \beta_h \, \text{CapRatio}_{i,t-1} + \Gamma_h' Z_{i,t-1} + \varepsilon_{i,t+h}, \tag{1}$$

where i indexes countries, t years, and $h \in \{0, \dots, 5\}$ horizons. The dependent variable is credit growth relative to GDP, measured under alternative windows (3-year and 5-year definitions) and reported both in annual and cumulative form. In the figure labels, $dlog_c 2g_3 y$ denotes cumulative 3-year credit-to-GDP growth, while $dlog_c 2g_5 y$ denotes cumulative 5-year credit-to-GDP growth. $\alpha_i^{(h)}$ are country fixed effects and $\gamma_t^{(h)}$ are global year fixed effects; $Z_{i,t-1}$ includes lagged macro controls used in the baseline and robustness designs.

Figure 1 summarizes the result visually. Across specifications, the estimated effect of capital on subsequent credit growth is generally positive: economies with stronger bank capital positions tend to experience larger subsequent credit-to-GDP expansions, although precision declines at longer horizons in tighter specifications. The identifying variation therefore comes from within-country fluctuations in capitalization after netting out common global shocks. We report three core specification setups: (i) country fixed effects only with HC1 inference (baseline-country-FE design), (ii) country plus year fixed effects with lagged controls and HC1 inference, and (iii) the same country-plus-year fixed-effects-with-controls specification with country-clustered standard errors.

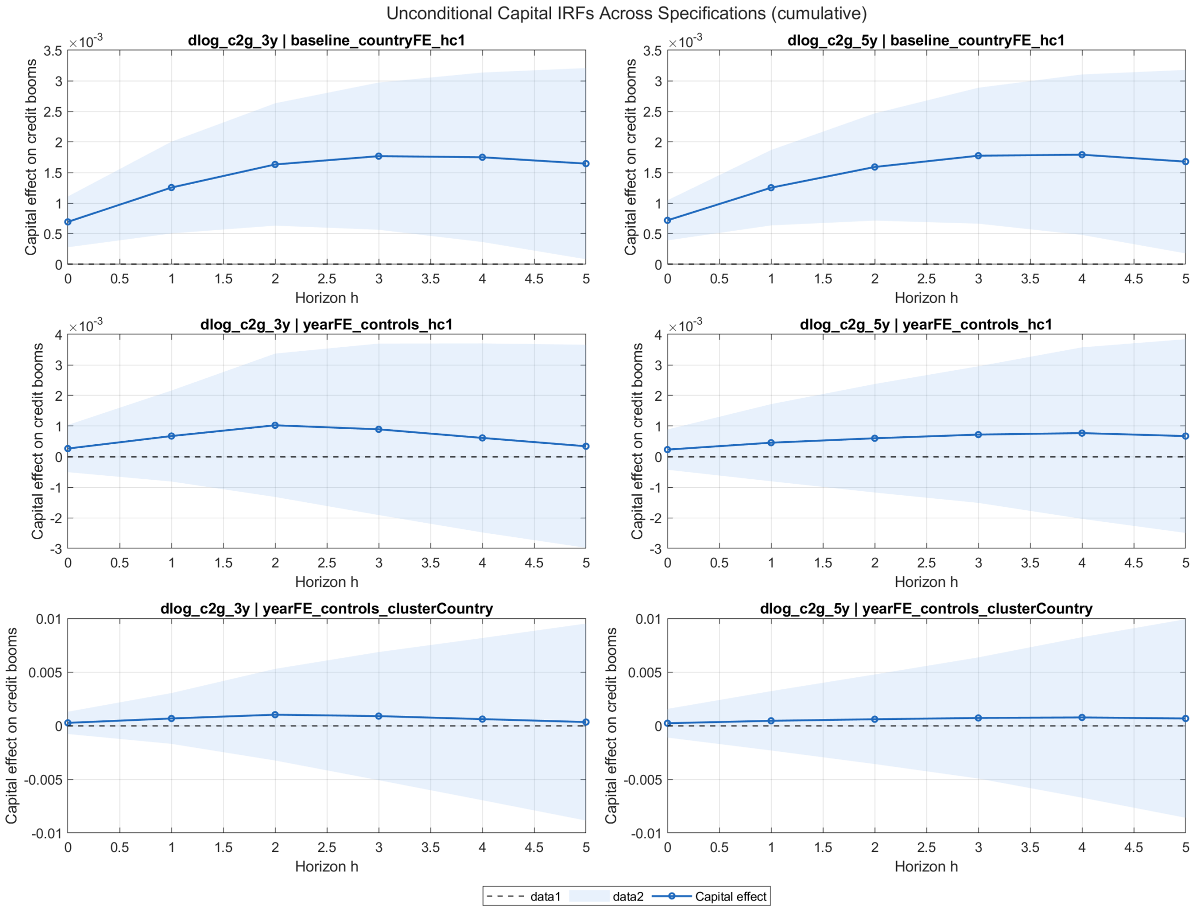

Figure 1: Panel LPs: cumulative response of credit-boom measures to lagged aggregate bank capital ratios

Notes: The figure overlays cumulative local-projection coefficients by horizon across alternative fixed-effects, controls, and inference specifications in the JST panel. The horizontal axis reports the horizon in years after the initial capital-ratio observation; the vertical axis reports the estimated response of credit-to-GDP growth. In the plot labels, $dlog_c2g3y$ denotes cumulative 3-year credit-to-GDP growth and $dlog_c2g5y$ denotes cumulative 5-year credit-to-GDP growth. The annual IRF counterpart is reported in Appendix A.

2.3 Credit booms and the output effects of negative productivity shocks

Fact 3: Negative productivity shocks are more contractionary when they hit after a credit boom. In the data, downturns that begin from high-credit-growth states are systematically deeper over subsequent years, indicating that prior balance-sheet expansion amplifies the real effects of adverse shocks. We estimate this pattern using interaction local projections.

$$\Delta_h y_{i,t+h} = \alpha_i^{(h)} + \gamma_t^{(h)} + \theta_h \mathbb{1}\{\Delta a_{i,t} < \tau\} + \delta_h \Big(\mathbb{1}\{\Delta a_{i,t} < \tau\} \times \text{Boom}_{i,t-1}\Big) + \Pi_h' X_{i,t-1} + u_{i,t+h}, \tag{2}$$

where $\Delta_h y_{i,t+h}$ is horizon-h output growth, $\mathbb{1}\{\Delta a_{i,t} < \tau\}$ identifies negative TFP shocks from PWT data, and $\text{Boom}_{i,t-1}$ is a lagged credit-boom indicator from the JST panel. We study two shock thresholds ($\tau = 0$ and lower-quintile TFP growth), annual and cumulative outcomes, and alternative boom definitions.

Figure 2 provides evidence in policy terms: when negative productivity shocks occur after a credit boom, output tends to fall by more over subsequent years than when the same shocks occur outside boom states. This is exactly the amplification pattern implied by the model's exposure channel.

As in Section 2.2, we report three core specification setups: baseline country fixed effects with HC1 inference, country-plus-year fixed effects with controls and HC1 inference, and country-plus-year fixed effects with controls and country-clustered inference. Controls include lagged macro-financial conditions to isolate differential amplification rather than average business-cycle variation. The interaction coefficient δ_h is the object of interest: it measures whether the output cost of a negative productivity shock is larger after a credit boom.

For boom definitions, we use high-credit-growth indicators based on the upper quantile of the pre-shock credit-to-GDP growth distribution, with alternative windows for credit accumulation (3-year and 5-year growth horizons). The baseline definition is the top-quintile 3-year credit-growth state; robustness replaces this with the corresponding top-quintile 5-year growth state.

Figure 2 reports cumulative responses under the $\tau = 0$ shock definition. Across baseline and robustness specifications, δ_h is negative over medium horizons, implying that output losses are deeper when adverse productivity shocks arrive in high-credit-boom states. This is the empirical counterpart of the model's credit-exposure channel.

Taken together, these results discipline the model by pinning down both the objects we calibrate and the mechanisms we emphasize. First, Fact 1 maps directly into the fragility hazard in Section 3.3: the logit evidence disciplines the slope and level of $\mathcal{H}(X)$ in Equation 6, so that at the sample-mean capital ratio of 15.5% the model reproduces a 5.8% annual pre-crisis probability, with a hazard slope consistent with the estimated marginal effect of capital on distress risk. Second, Facts 2 and 3 discipline the model's credit-exposure mechanism. Fact 2 supports the idea that stronger bank balance sheets

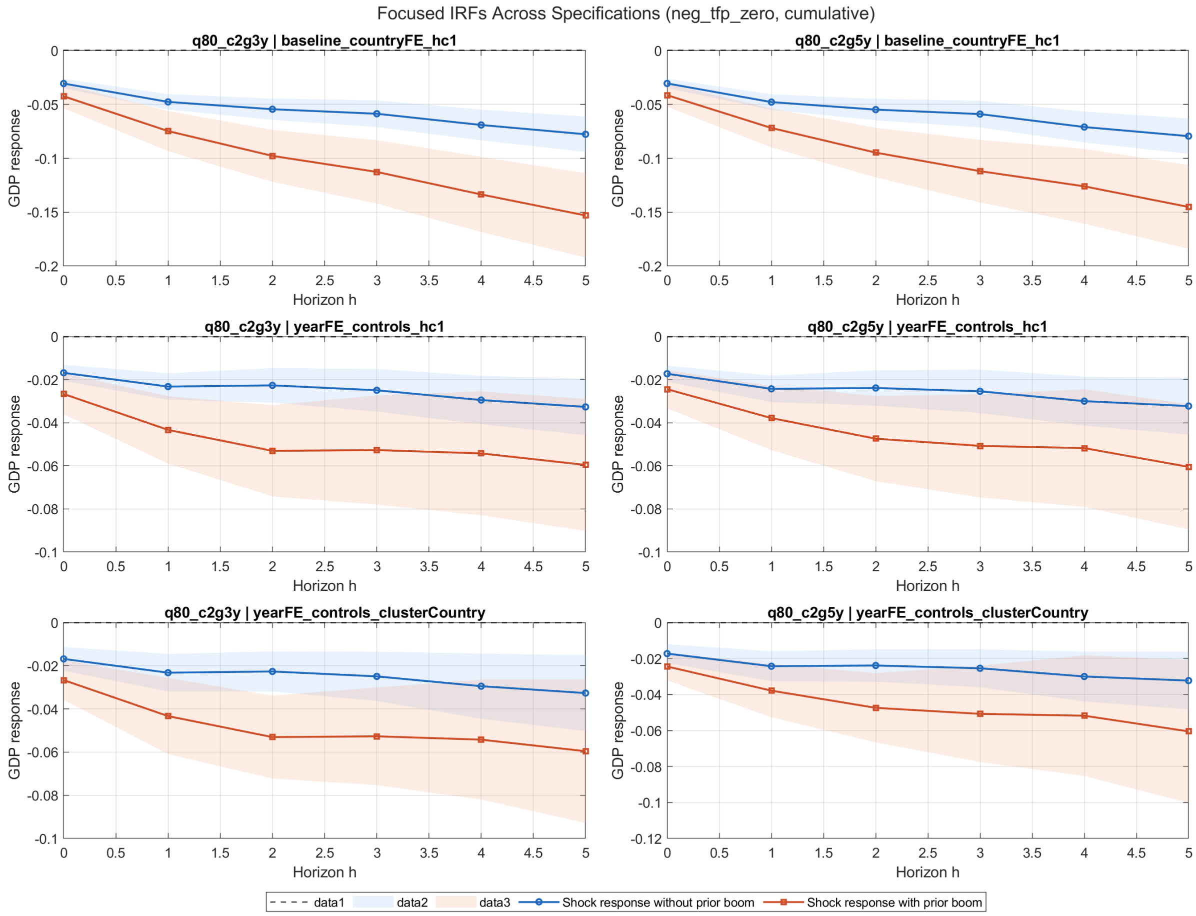

Figure 2: Panel LPs: cumulative output response to negative TFP shocks, conditional on prior credit-boom state

Notes: The figure plots interaction coefficients between negative TFP shocks and lagged credit-boom states across specification variants. The horizontal axis reports the horizon in years after the shock; the vertical axis reports the additional cumulative output response associated with being in a prior credit-boom state. Negative values mean that output falls more after adverse productivity shocks when thay follow a credit-boom. TFP denotes total factor productivity. The three alternative IRF variants are reported in Appendix A.

permit larger balance-sheet expansion, which in the model operates through the bank's lending choice and the leverage constraint in Equations 4 and 5. Fact 3 then disciplines the quantitative importance of that mechanism by showing that downturns are more severe after credit booms, matching the model's prediction that high lending states amplify negative productivity shocks through the same intermediation channel. In this sense, the empirical section does not just motivate the model qualitatively: it anchors the hazard block and identifies the balance-sheet channel that drives the model's state-dependent amplification results in Sections 4 and 5.

3 Model

Before turning to the formal structure, the model can be understood in simple terms as combining two opposing effects of bank equity. On one side, low equity makes banks fragile: when the banking sector is highly leveraged, the probability of a failure, run, or funding stress rises, so scarce equity amplifies financial crises. This is the fragility channel. On the other side, high equity relaxes banks' balance-sheet constraints and supports more lending in good times; that larger credit exposure raises output when fundamentals are strong but makes the real economy more sensitive when adverse productivity shocks arrive. This is the credit-exposure channel. The capitalization dilemma comes from the fact that the same policy lever - bank capital - moves both forces at once: more equity insures the economy against financial crises, but it can also increase exposure to real-sector downturns by enabling larger balance sheets before the shock.

3.1 Environment

Time is discrete and infinite. The economy consists of a representative household, a unit continuum of atomistic banks, and a representative firm. The aggregate state is $X = [N, A]$, where N is aggregate bank equity and A is aggregate TFP. We use lowercase letters (e.g., n, l, b) for individual bank variables. Each bank is of measure zero and takes the aggregate capital ratio N/L—and hence the failure hazard $\mathcal{H}(X)$—as given; we study the symmetric equilibrium, in which individual choices coincide with their aggregate counterparts ($n = N$, $l = L$, $b = B$). Because each atomistic bank ignores its own contribution to the aggregate N/L, the decentralized equilibrium is inefficient—the source of the externality the planner corrects in Section 3.7.

3.2 Bank

The bank maximizes its franchise value:

$$\begin{aligned} J(n;X) = \max_{n'} \; & \pi(n;X) + n - n' \\ & + (1-\mathcal{H}(X))\,\mathbb{E}\,q(X,X')\,J(n';X') \\ & + \mathcal{H}(X)\,\mathbb{E}\,q(X,X')\,J\big((1-\varphi)n';X'\big) \end{aligned} \tag{3}$$

where $\pi(n;X)$ denotes bank profit, $q(X,X')$ is the stochastic discount factor, $\mathcal{H}(X)$ is the endogenous fragility hazard rate, and φ is the fraction of equity destroyed in a failure. The two continuation values embed the dual nature of the bank's environment: with probability $1 - \mathcal{H}$, the bank enters the next period intact; with probability $\mathcal{H}$, it suffers an equity loss that degrades its future intermediation capacity.

Bank profit is:

$$\pi(n; X) = \max_{l} \ \Phi(l)(R^K(X) - 1) - (R^B(X) - 1)(l - n) \tag{4}$$

where $\Phi(l) = \Phi_A l^{\Phi_B}$ is a concave intermediation technology ($\Phi_B < 1$), R^K is the gross return on capital, R^B is the gross deposit rate, and l is total lending. Deposits are $b = l - n$. The concavity captures diminishing returns in intermediation: the bank lends l units, which are transformed into $\Phi(l) \leq l$ units of productive capital. The gap $l - \Phi(l)$ represents the resource cost of intermediation.

The bank faces a leverage constraint:

$$b \leq \phi_0 + \phi_1 n \tag{5}$$

with associated multiplier $\lambda \geq 0$.

Intuitively, this constraint turns equity losses into a nonlinear lending response. When the constraint is slack, a small decline in bank equity can be absorbed mostly through balance-sheet adjustment; once the constraint binds, the same loss forces deposits and lending to contract one-for-one, creating the kink that later appears as the economy's "fault line".

3.3 Fragility hazard

The failure probability is:

$$\mathcal{H}(X) = \sigma\left(\kappa_1 \frac{N}{L(X)} + \kappa_0\right) \tag{6}$$

where $\sigma(\cdot)$ is the logistic sigmoid, N/L is the aggregate capital ratio, and $\kappa_1 < 0$ so that lower capitalization raises fragility. Thus, a lower aggregate capital ratio N/L - i.e., a more highly leveraged banking system - sharply raises the probability of a crisis, consistent with Fact 1. Intuitively, a thinly capitalized banking sector has less capacity to absorb losses and is more vulnerable to depositor withdrawals, asset fire sales, or interbank contagion—all of which are captured in reduced form through the capital ratio. We use a logistic hazard because it is parsimonious, maps naturally into probabilities bounded between zero and one, and can be disciplined directly by the logit evidence in Section 2.1. Alternative monotone hazard forms would preserve the qualitative mechanism as long as fragility rises steeply when capitalization falls; the logistic specification provides the most direct empirical mapping while keeping the global solution tractable. The parameter κ_1 is calibrated from a logit regression of crisis probability on the capital ratio using the Jordà et al. (2017) macrohistory database.

3.4 Firm

A representative firm operates a Cobb-Douglas technology:

$$Y = AK^{\alpha}(H^D)^{1-\alpha} \tag{7}$$

and hires capital and labor competitively. Setting $\delta = 1$ (full depreciation) eliminates capital as an endogenous state variable, leaving bank equity as the sole aggregate endogenous state. This preserves the key economic forces while maintaining tractability; we discuss relaxing this assumption in Section 6.

3.5 Household

The representative household has GHH preferences with a general CRRA parameter γ:

$$u(C, H^S) = \frac{\left(C - \eta \frac{(H^S)^{1+1/\chi}}{1+1/\chi}\right)^{1-\gamma}}{1-\gamma} \tag{8}$$

with discount factor β, Frisch elasticity χ, and labor disutility weight η. In the baseline calibration, $\gamma = 1$, so the utility function reduces to $u = \log\big(C - \eta (H^S)^{1+1/\chi}/(1+1/\chi)\big)$.

The household budget constraint is:

$$C + p'(X)\, a' = p(X)\, a + \big(R^B(X) - 1\big)\, B + w(X)\, H^S - \xi(B) \tag{9}$$

where $\xi(B) = \zeta_1 B + \zeta_2 B^2$ is a portfolio adjustment cost for deposits, $p(X) = J(N; X)$ is the bank equity price, and a is equity holdings. Deposits are intra-period: the household earns the within-period spread $(R^B(X) - 1)\, B$ on funds placed with the banking sector, so only $\xi(B)$ is a real resource cost (in equilibrium, with $a = 1$, this is the consumption form $C = (R^B(X) - 1)\, B + d + w(X)\, H^S - \xi(B)$ used in the solution, where d is the bank dividend). When a failure occurs (probability $\mathcal{H}(X)$), a fraction φ of bank equity is destroyed, mapping $N \mapsto (1 - \varphi)\, n'(N; X)$; the lost equity is not rebated, so bank failures destroy aggregate resources.

3.6 Equilibrium

Definition 1 (Recursive Competitive Equilibrium). *A recursive competitive equilibrium is a collection of: (i) a bank value function $J(n;X)$ and policy functions $\{n'(n;X), l(n;X)\}$; (ii) a household value function $v(a;X)$ and policy functions $\{a'(a;X), B(a;X), H^S(a;X)\}$; (iii) pricing functions $\{R^K(X), R^B(X), w(X), p(X), p'(X), q(X,X')\}$; and (iv) an aggregate law of motion $N' = \Gamma(X,\varepsilon)$; such that: (a) the bank's and household's policy functions solve their respective Bellman equations; (b) all markets clear ($\Phi(L) = K$, $B(1;X) = l - n$, $a(1;X) = 1$, $H^S = H^D$); (c) the aggregate law of motion is consistent with individual decisions; and (d) prices are consistent with optimality.*

In equilibrium, all markets clear. The key aggregate conditions are:

- *Equity transition.* Aggregate bank equity evolves according to

$$N'(X) = \begin{cases} n'(N;X) & \text{with prob. } 1-\mathcal{H}(X) \\ (1-\varphi)\, n'(N;X) & \text{with prob. } \mathcal{H}(X) \end{cases} \tag{10}$$

- *National accounting.* The national accounting identity is

$$Y = C + \xi(B) + K + (N' - N) \tag{11}$$

  where C is consumption, $\xi(B)$ is a real resource cost (deadweight loss from intermediation), and bank dividends $d = \pi + n - n'$ flow to households as equity income.

- *Stochastic discount factor.* The SDF is

$$q(X,X') = \beta \frac{u_c(C', H^{S'})}{u_c(C, H^S)} = \beta \frac{\widetilde{C}(X)}{\widetilde{C}(X')} \tag{12}$$

  where $\widetilde{C} := C - \eta (H^S)^{1+1/\chi}/(1+1/\chi)$ is composite consumption and the second equality holds under log-GHH preferences ($\gamma = 1$, so $u_c = 1/\widetilde{C}$).

3.7 Social planner's problem

The constrained social planner (SPP) solves the same problem as the bank but internalizes the effect of aggregate equity on the hazard rate $\mathcal{H}(X)$. The bank's first-order condition for n' (the Euler equation) is:

$$1 = (1-\mathcal{H})\, \mathbb{E}\left[q\, J_1(n';X')\right] + \mathcal{H}\,(1-\varphi)\, \mathbb{E}\left[q\, J_1\big((1-\varphi)n';X'\big)\right] \tag{13}$$

where the envelope condition gives $J_1(n; X) = R^B(X) + \lambda(1 + \phi_1)$. The planner modifies this condition by adding a Pigouvian wedge τ that captures the systemic benefit of higher capitalization:

$$J_1^{\text{SPP}}(n; X) = R^B(X) + \lambda(1 + \phi_1) + \tau(X) \tag{14}$$

where the Pigouvian tax is:

$$\tau(X) = \frac{\beta}{u_c(C, H^S)} \cdot \frac{\partial \mathcal{H}}{\partial B} \cdot \mathbb{E}\left[V(n'; X') - V\big((1 - \varphi)n'; X'\big)\right] \tag{15}$$

Here $V(N; X) := v(1; [N, A])$ is the household's equilibrium lifetime utility as a function of the aggregate state, and $\partial \mathcal{H}/\partial B > 0$ reflects that higher leverage increases the failure hazard (since $\partial(N/L)/\partial B = -N/L^2 < 0$ and $\kappa_1 < 0$). Because $V(n') > V((1 - \varphi)n')$, the tax $\tau > 0$: the planner penalizes leverage beyond what private banks choose. The underlying externality is compositional: each bank takes the aggregate failure probability $\mathcal{H}(X)$ as given, ignoring that its own leverage choice contributes to the sector-wide capital ratio that determines $\mathcal{H}$. The spillover operates through the aggregate quantity N/L rather than a relative price, distinguishing it from the pecuniary externalities of the collateral-constraint literature. The modified lending FOC becomes $\Phi'(l)(R^K - 1) - (R^B - 1) = \lambda + \tau$, inducing lower leverage and higher equity accumulation.

3.8 Analytical properties

The following propositions characterize key properties of the equilibrium. Proofs are provided in Online Appendix B.

Proposition 1 (Sign of the Pigouvian wedge)**.**
In the constrained social planner's problem, the Pigouvian wedge satisfies $\tau(X) > 0$ for all X with $\mathcal{H}(X) \in (0, 1)$. The planner accumulates strictly more equity than the competitive equilibrium: $n'^{\text{SPP}}(N; X) > n'^{\text{CE}}(N; X)$.

Proof. See Online Appendix B. □

The result follows from two observations: (i) $\partial \mathcal{H}/\partial B > 0$, since $\kappa_1 < 0$ and $\partial(N/L)/\partial B = -N/L^2 < 0$; and (ii) $V(n'; X') > V((1 - \varphi)n'; X')$ for $\varphi > 0$. Since $\tau > 0$ enters as $J_1^{\text{SPP}} = R^B + \lambda(1 + \phi_1) + \tau > J_1^{\text{CE}}$, the planner's Euler equation requires higher equity retention.

Proposition 2 (Conditional steady state).
Fix $A \in \{A_L, A_H\}$. If the equilibrium policy function $n'(\cdot\,; A)$ is continuous with $n'(N; A) > N$ for N sufficiently small and $n'(N; A) < N$ for N sufficiently large, then a conditional steady state N_A^{CS} exists. If additionally $n'(\cdot\,; A)$ crosses the 45-degree line exactly once, the CSS is unique and globally stable under the frozen regime.

Proof. See Online Appendix B. □

The CSS captures the equity level at which dividend payouts exactly offset the bank's retained earnings motive: below N_A^{CS}, equity is scarce and intermediation profits induce accumulation; above it, the marginal return falls below the household's required rate of return and equity is paid out.

Proposition 3 (Planner's CSS ordering).
Under the conditions of Proposition 2, $N_A^{\mathrm{CS,SPP}} > N_A^{\mathrm{CS,CE}}$ for each TFP regime A.

Proof. See Online Appendix B. □

This follows directly from Proposition 1: since the planner retains more equity at every N, the policy function $n'^{\mathrm{SPP}}(\cdot; A)$ lies above $n'^{\mathrm{CE}}(\cdot; A)$, shifting the fixed point rightward.

4 Quantitative analysis

4.1 Calibration

Table 2 summarizes the baseline calibration. Household preferences follow standard values: discount factor $\beta = 0.96$ (annual), log utility ($\gamma = 1$), unit Frisch elasticity ($\chi = 1$), and labor disutility $\eta = 0.65$ to target steady-state labor supply near one-third. The capital share is $\alpha = 0.33$.

For banking parameters, the intermediation curvature $\Phi_B = 0.9$ generates moderate diminishing returns. The quadratic deposit cost $\zeta_2 = 0.5$ pins down the deposit rate spread; the linear component $\zeta_1 = 0$ is set to zero for parsimony. The leverage parameters $\phi_0 = 0.01$ and $\phi_1 = 1.5$ are chosen so that the constraint binds approximately 20% of the time in the CE.[1]

[1] The implied steady-state capital ratio $N/L \approx 0.50$ is higher than typical bank leverage ratios (5–10%). This is a direct consequence of full depreciation ($\delta = 1$): the entire capital stock is rebuilt each period, so the bank's balance sheet L reflects only one period's lending flow. Under realistic annual depreciation ($\delta \approx 0.10$), the steady-state capital stock $K_{\mathrm{ss}} = I_{\mathrm{ss}}/\delta$ is an order of magnitude larger for the same output level. Since the intermediation technology $K = \Phi(L)$ links capital to lending, a larger durable K requires a proportionally larger L, inflating the bank's balance sheet. Bank equity N, however, is determined by the

The hazard slope $\kappa_1 = -6.86$ is estimated from a logit regression of annual crisis probability on the capital ratio using the Jordà et al. (2017) macrohistory database (the empirical logit intercept is -1.72). Only the slope is taken directly from the data; the model intercept $\kappa_0 = 0.47$ is recalibrated so that the hazard evaluated at the model's steady-state ratio $N/L \approx 0.50$ matches a reference steady-state hazard of about 5% (the empirical crisis probability at the mean capital ratio, 5.8%). The model's *ergodic* crisis frequency is higher (8.9% in the CE, Table 4), since the stationary distribution spends time below the steady-state ratio. The distress loss $\varphi = 0.60$ implies 60% equity destruction upon failure. We set the equity loss in distress to $\varphi = 0.60$ to represent a severe but not total banking sector impairment: the shock destroys a majority of bank equity while allowing the intermediary sector to continue operating, consistent with the model's interpretation of distress as a run, funding stress, or resolution event rather than liquidation.

4.2 Solution method

We solve the model using the Repeated Transition Method (RTM) of Lee (2025). The key computational challenge is tracing counterfactual paths to evaluate the conditional expectations in the Euler equation (13) accurately. The algorithm proceeds as follows:

1. *Backward step.* Given aggregate allocations, the bank's Euler equation (13) is solved for the optimal equity choice $n'(N; X)$ using iso-shock matching, simultaneously determining the household's labor supply and the deposit market equilibrium.

2. *Forward step.* The economy is simulated using the resulting policy functions and aggregate allocations are updated via damped iteration.

3. *Convergence.* The two steps alternate until the mean squared error between successive aggregate allocations falls below 10^{-8}.

The simulation uses $T = 5{,}001$ periods with a burn-in of 500. The deterministic steady state (computed under $\mathcal{H} = 0$) serves as the initial guess; the stochastic equilibrium endogenously determines positive hazard rates.

4.3 Conditional saddle paths

Following Lee (2026), we analyze the model's equilibrium dynamics through *conditional saddle paths*—the stochastic analogue of deterministic phase diagrams. Fix a TFP regime

intertemporal optimization (the Euler equation) and does not scale with the capital stock. The result is a much lower N/L under partial depreciation. The leverage parameters are calibrated to match a binding frequency target (20%) rather than a capital ratio level.

Table 2: Baseline Calibration

Parameter	Description	Value
Households		
β	Discount factor	0.960
γ	CRRA coefficient	1.000
η	Labor disutility weight	0.650
χ	Frisch elasticity	1.000
Firms		
α	Capital share	0.330
δ	Depreciation rate	1.000
Banking		
Φ_A	Intermediation scale	1.000
Φ_B	Intermediation curvature	0.900
ζ_1	Deposit cost (linear)	0.000
ζ_2	Deposit cost (quadratic)	0.500
ϕ_0	Leverage intercept	0.010
ϕ_1	Leverage slope	1.500
Fragility		
κ_1	Hazard slope (capital ratio)	-6.86
κ_0	Hazard intercept	0.47
φ	Distress recovery loss	0.60
Aggregate TFP shock		
A	TFP grid	$\{0.98,\ 1.02\}$
Π_A	TFP transition	$\begin{pmatrix} 0.90 & 0.10 \\ 0.10 & 0.90 \end{pmatrix}$

Notes: Lending technology: $\Phi(l) = \Phi_A\, l^{\Phi_B}$. Deposit cost: $\xi(B) = \zeta_1 B + \zeta_2 B^2$. Leverage: $B \leq \phi_0 + \phi_1 N$. Hazard: $\mathcal{H} = \sigma(\kappa_1\, N/L + \kappa_0)$, where $\sigma(\cdot)$ is the logistic sigmoid and $N/L = N/(N+B)$. κ_1 from Jordà et al. (2017) logit (annual, capital ratio).

$A \in \{A_L, A_H\}$ and an initial equity level N_0. The *conditional saddle path* $\mathcal{M}(A;\, N_0) := \{(N_t, C_t)\}_{t\geq 0}$ is the orbit generated by the policy function $n'(N_t;\, A)$ and the consumption mapping $C(N_t, A)$ under the frozen regime A—no TFP switches, no bank failures. Crucially, these objects are computed from the decision rules of the *stochastic* equilibrium: agents correctly anticipate future regime switches and bank failures, so the saddle paths embed forward-looking expectations even though the exogenous state is held fixed. The limit point $(N_A^{\mathrm{CS}}, C_A^{\mathrm{CS}})$, the fixed point of $n'(\cdot\,;A)$, is the *conditional steady state* (CSS). Table 3 reports CSS values; Figure 3 plots the saddle paths.

The conditional saddle paths decompose all equilibrium fluctuations into two types of movements. *Across-saddle* jumps occur when the exogenous state switches: since bank

Table 3: Conditional Steady States

		N_{css}	C_{css}
CE (Competitive)	$A = A_L$ (Recession)	0.0305	0.2128
	$A = A_H$ (Expansion)	0.0424	0.2400
SPP (Social Planner)	$A = A_L$ (Recession)	0.0537	0.2130
	$A = A_H$ (Expansion)	0.0721	0.2429

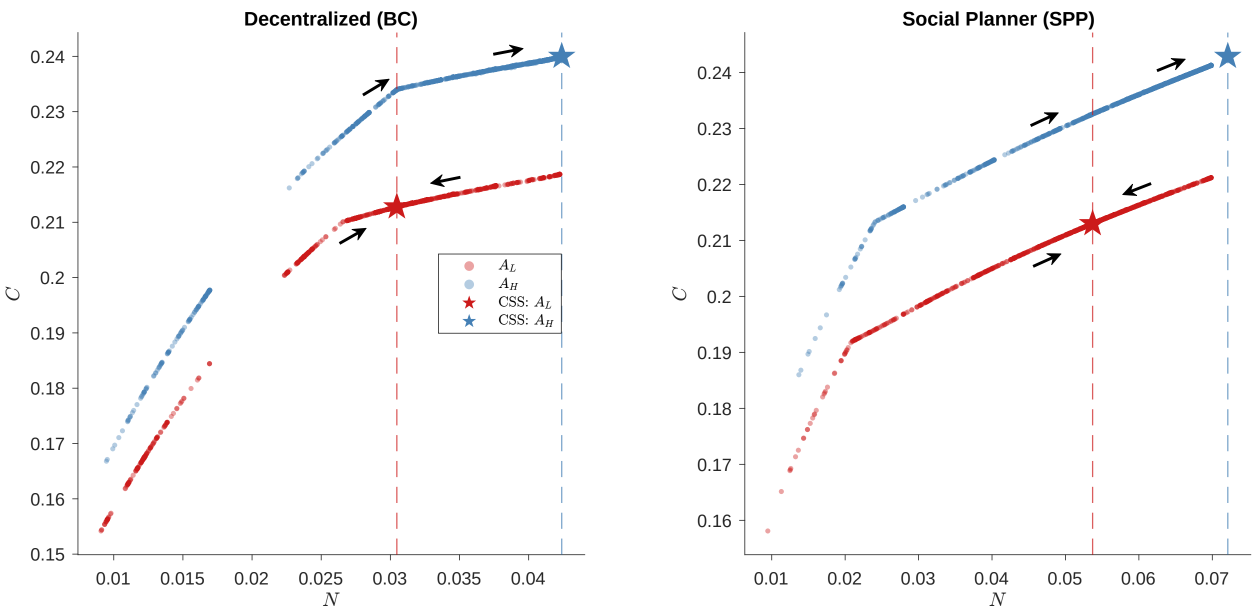

Figure 3: Conditional saddle paths: CE vs. SPP.

Notes: Conditional saddle paths under frozen TFP regimes, computed from the globally solved model. Stars mark conditional steady states. Panel (a): CE. Panel (b): SPP. In the SPP, the planner's recession CSS ($N = 0.0537$) exceeds the CE expansion CSS ($N = 0.0424$).

equity N is predetermined, a TFP switch from A_H to A_L corresponds to a vertical jump from the expansion saddle to the recession saddle at the same N. *Along-saddle* movements capture the endogenous propagation that follows: once on a given saddle, the economy converges toward the corresponding CSS through the equilibrium policy function. Together, these two forces—exogenous transitions across saddles and endogenous propagation along them—trace out the full stochastic equilibrium path in the (N, C) plane.

The across-saddle gap measures the *credit exposure channel*. The vertical distance

$$\Delta C(N) \;:=\; C(N, A_H) - C(N, A_L) \tag{16}$$

gives the impact effect of a TFP switch on consumption. Because the expansion saddle is steeper—higher TFP amplifies the marginal return to bank-intermediated credit—$\Delta C(N)$ widens with N. This generates both size and duration dependence: longer booms push N toward $N^{\text{CS}}_{A_H}$, widening the gap at the point of reversal. Geometrically, the slope difference between the two saddles is the primitive source of state-dependent amplification: if the saddles were parallel, TFP shocks would produce identical consumption drops regardless of capitalization.

The along-saddle dynamics encode the *fragility buffer channel*. A bank failure displaces the economy leftward along the active saddle path by destroying a fraction φ of equity. How much this displacement costs depends on the saddle's local curvature. The occasionally binding leverage constraint (5) creates a kink at intermediate equity levels, dividing the state space into two regimes. Above the kink, equity is abundant relative to the constraint, deposits adjust freely, and losses are absorbed smoothly—the economy operates as if unconstrained. Below the kink, the constraint binds, deposits are rationed, and each unit of lost equity forces a contraction in lending that propagates to output: a credit crunch. The kink is the economy's "fault line"—crossing it transforms a manageable equity loss into a self-reinforcing deleveraging spiral. In the CE, the constraint binds roughly 20% of the time. The planner shifts both saddle paths rightward (Table 3), keeping the economy above the kink for the vast majority of periods (binding only 2% of the time) and neutralizing this amplification channel.

4.4 Generalized impulse response functions

We compute state-dependent generalized impulse response functions (GIRFs; Koop et al., 1996):

$$\text{GIRF}_t = \mathbb{E}[\text{path}_t \mid \text{shock},\ N_0] - \mathbb{E}[\text{path}_t \mid \text{no shock},\ N_0] \tag{17}$$

expressed as a percentage of steady-state values, with expectations over 1,000 Monte Carlo paths. We condition on N_0 at the 10th percentile (fragile) and 90th percentile (well-capitalized) of the ergodic distribution.

4.4.1 TFP shock

The key message from the TFP experiment is that strong bank balance sheets are a double-edged sword. High equity lowers fragility, but it also supports a larger intermediation volume, so a negative productivity shock hits a more exposed banking system and produces a larger output decline. Figure 4 confirms that well-capitalized banks amplify TFP shocks: starting from high N, the proportional output decline is roughly twice as large as from low N (-10% vs. -6% at impact). The mechanism is a credit multiplier operating in reverse—the very lending that boosts output in good times magnifies the contraction in bad times. The GIRF compares paths starting in recession (A_L) against expansion (A_H), expressed as the percentage difference relative to the counterfactual (no-shock) path: $100 \times (\text{shocked} - \text{unshocked})/\text{unshocked}$.[2]

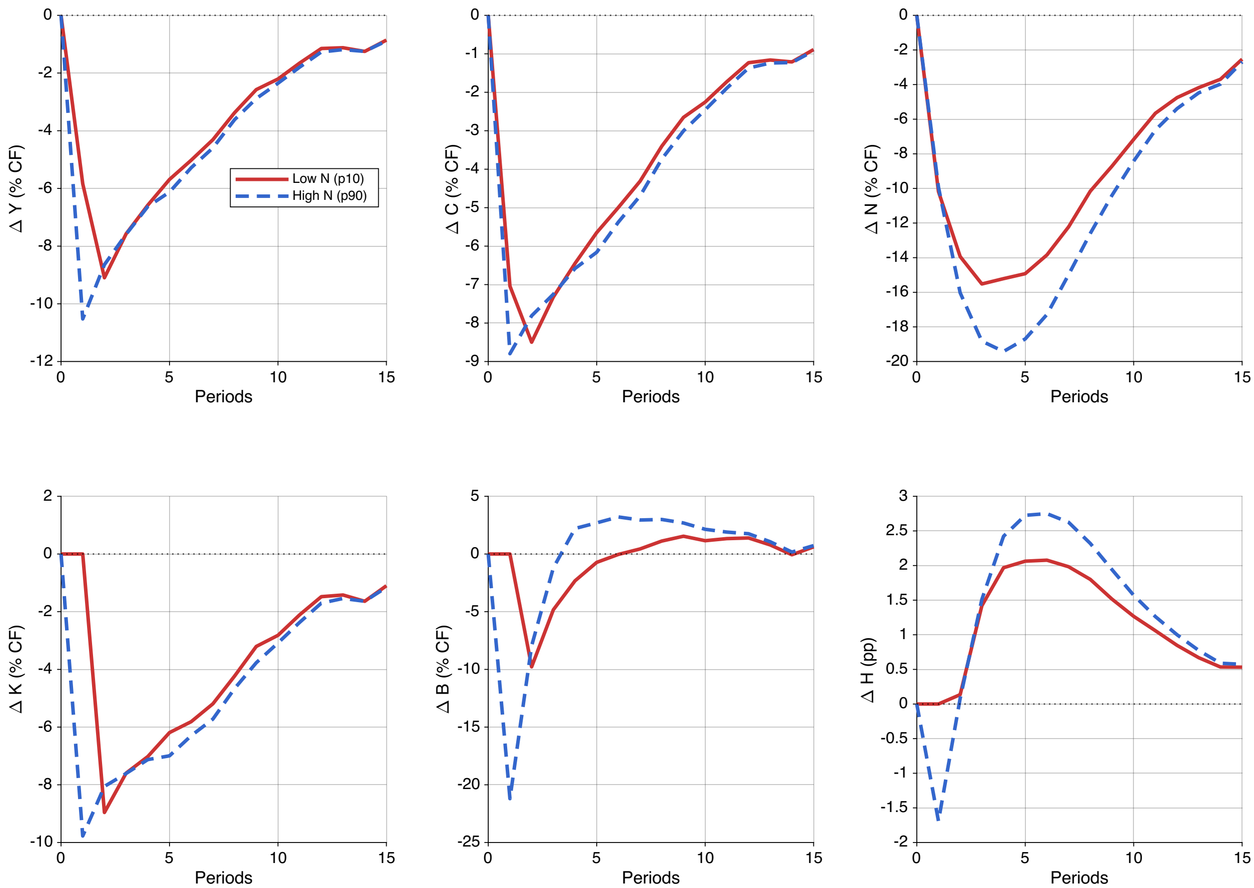

Figure 4: State-dependent GIRFs: negative TFP shock.

Notes: GIRF expressed as percentage difference relative to the counterfactual (no-shock) path. Red solid: low N_0 (p10); blue dashed: high N_0 (p90).

4.4.2 Bank hazard shock

The key message from the hazard-shock experiment is the opposite of the TFP case: low capitalization dramatically worsens crisis outcomes. The shock mechanically destroys

[2] The two paths draw independently from the Markov transition matrix for future TFP but share uniform draws for run realizations to reduce Monte Carlo variance. The counterfactual normalization ensures that the GIRF captures the *proportional* impact of the shock, removing level differences across conditioning states.

the same fraction of bank equity in every conditioning state, but its macroeconomic cost depends on whether the economy starts safely above the leverage kink or close to the fault line. Figure 5 shows fragility amplification during expansion: starting from low N, the proportional output decline (-30%) is far deeper than from high N (-5%), and the recovery is slow because the equity loss pushes the economy below the leverage kink, triggering a credit crunch. Starting from high N, the same proportional equity loss is absorbed within the unconstrained region, and the economy recovers within a few periods. Figure 6 shows that damage is smaller in recession, as banks are already partially deleveraged and closer to the kink regardless of initial capitalization. The GIRF isolates this financial channel by forcing a bank failure at $t = 1$ ($\varphi = 60\%$ equity loss), while both paths share the same TFP sequence; from $t \geq 2$, failures are drawn endogenously with shared uniform draws. Under the counterfactual normalization, the equity panel shows a uniform -60% initial drop across all conditioning states, so the figure isolates *recovery dynamics* rather than the mechanical shock itself.

4.4.3 CE vs. SPP

We now overlay the CE and SPP impulse responses to isolate the planner's fragility correction. The plain-language trade-off is this: the planner's crisis-prevention policy makes banks hold more capital in every state, but healthier banks can swing more. As a result, a negative TFP shock causes a slightly larger contraction under the planner when starting from low N - about 9% rather than 6% - which is the cost of the extra safety gained against crises.

Figure 7 compares TFP shock responses, conditioning both CE and SPP on the *same* initial equity levels (the CE's 10th and 90th percentiles) to ensure a like-for-like comparison.[3] Two features emerge. First, in output and consumption (Y, C panels), the CE and SPP responses are similar but not identical: at low N, the SPP's proportional output drop ($\approx -9\%$) is slightly larger than the CE's ($\approx -6\%$). This is the capitalization dilemma in its precise form: at the same equity level, the planner's allocation creates *more* proportional TFP exposure. The mechanism is not that the SPP intermediates more—it actually intermediates slightly less (lower B, lower L)—but that the planner's Pigouvian tax replaces the binding leverage constraint with a smooth price, giving the bank more freedom to adjust its balance sheet in response to TFP. In the CE, the binding constraint acts as a straitjacket that limits both upside and downside; the SPP's voluntary restraint unravels

[3] If each economy is conditioned on its own percentiles, level differences between CE and SPP confound the comparison. Conditioning on common initial N isolates the effect of the planner's policy function at the same starting point.

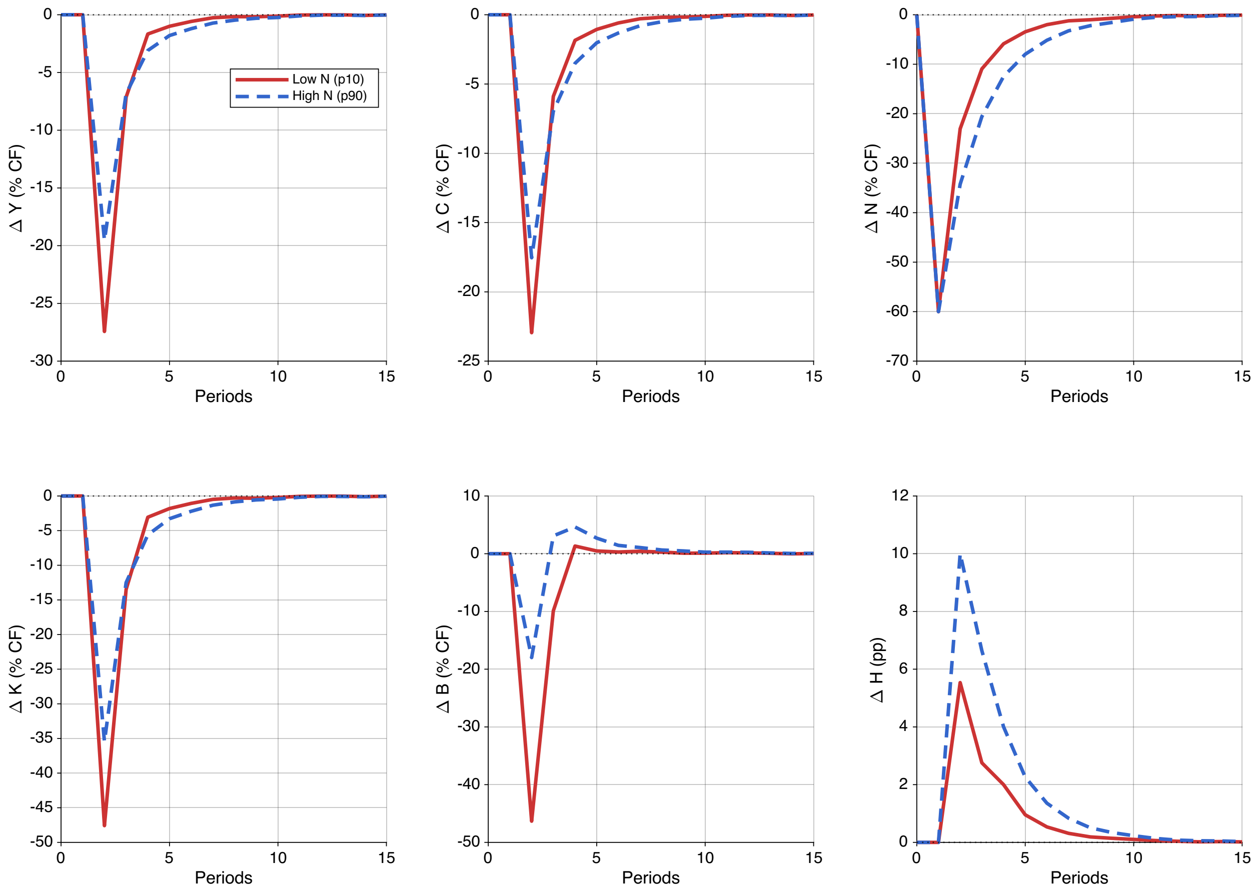

Figure 5: State-dependent GIRFs: bank hazard shock (expansion).

Notes: Both paths share the same TFP sequence starting in expansion. The shock forces a bank failure at $t = 1$. Red solid: low N_0 (p10); blue dashed: high N_0 (p90). GIRF expressed as percentage difference relative to the counterfactual (no-shock) path.

during stress, producing wider swings.

Second, the deposit panel (ΔB) reveals a sharp CE–SPP asymmetry. At both conditioning states, the SPP's deposits respond more dramatically than the CE's—rising in relative terms as the planner's unconstrained bank adjusts freely, while the CE bank's deposits are pinned by the binding constraint. This flexibility is the mirror image of the fragility benefit: the same freedom that allows the planner to stabilize fragility makes the economy more responsive to TFP.

The contrast is stark for financial shocks. Figure 8 overlays the expansion-state hazard shock responses. Under the counterfactual normalization, the equity panel shows a uniform -60% initial drop for all four lines—the mechanical destruction is identical—so the figure isolates the recovery dynamics. In the CE, the fragile bank (low N_0) suffers a deep, persistent proportional output contraction (-30%) as the leverage constraint binds and deleveraging amplifies the initial equity destruction. The planner's economy, starting

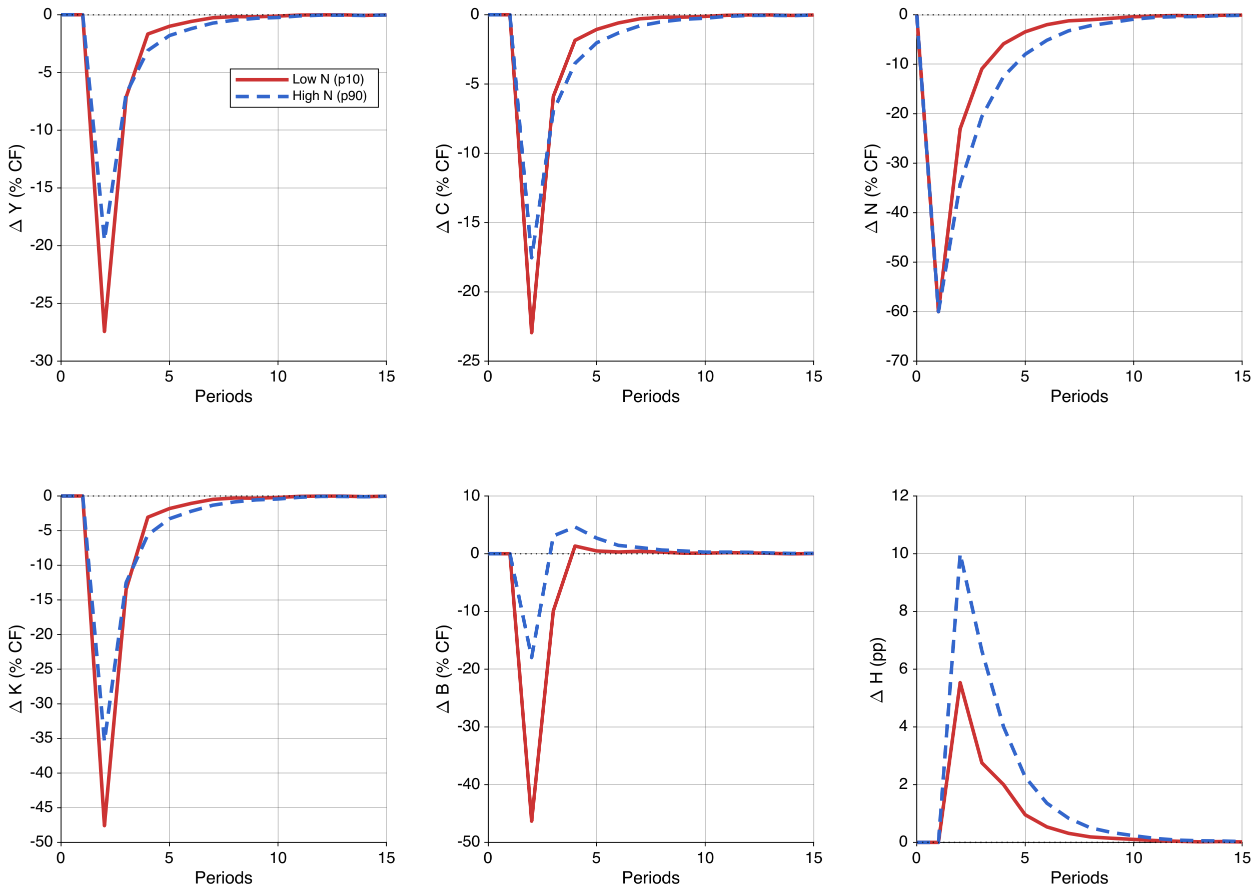

Figure 6: State-dependent GIRFs: bank hazard shock (recession).

Notes: Same construction as Figure 5, with both paths starting in recession.

from the same N_0, absorbs the same proportional equity loss but recovers much faster: the constraint remains slack (because the Pigouvian tax has already induced lower leverage), so the bank can rebuild its balance sheet without hitting the hard cap. The deposit panel is particularly striking: CE deposits collapse (-70%), while SPP deposits *increase* ($+80\%$ at high N) as the unconstrained bank expands intermediation to exploit the post-crisis recovery.

Figure 9 shows that this pattern persists in recession, though the CE–SPP gap narrows. In recession, lower TFP compresses intermediation profits and both economies have drawn down equity, so even the CE bank is closer to the constrained region—the planner's insurance margin shrinks. Taken together, the three comparisons reveal the capitalization dilemma in its full form. The planner's Pigouvian correction replaces the rigid leverage constraint with a flexible price, which is decisive for financial shocks (the unconstrained bank recovers faster and avoids the credit crunch) but creates wider proportional swings in response to TFP shocks (the unconstrained bank adjusts more freely

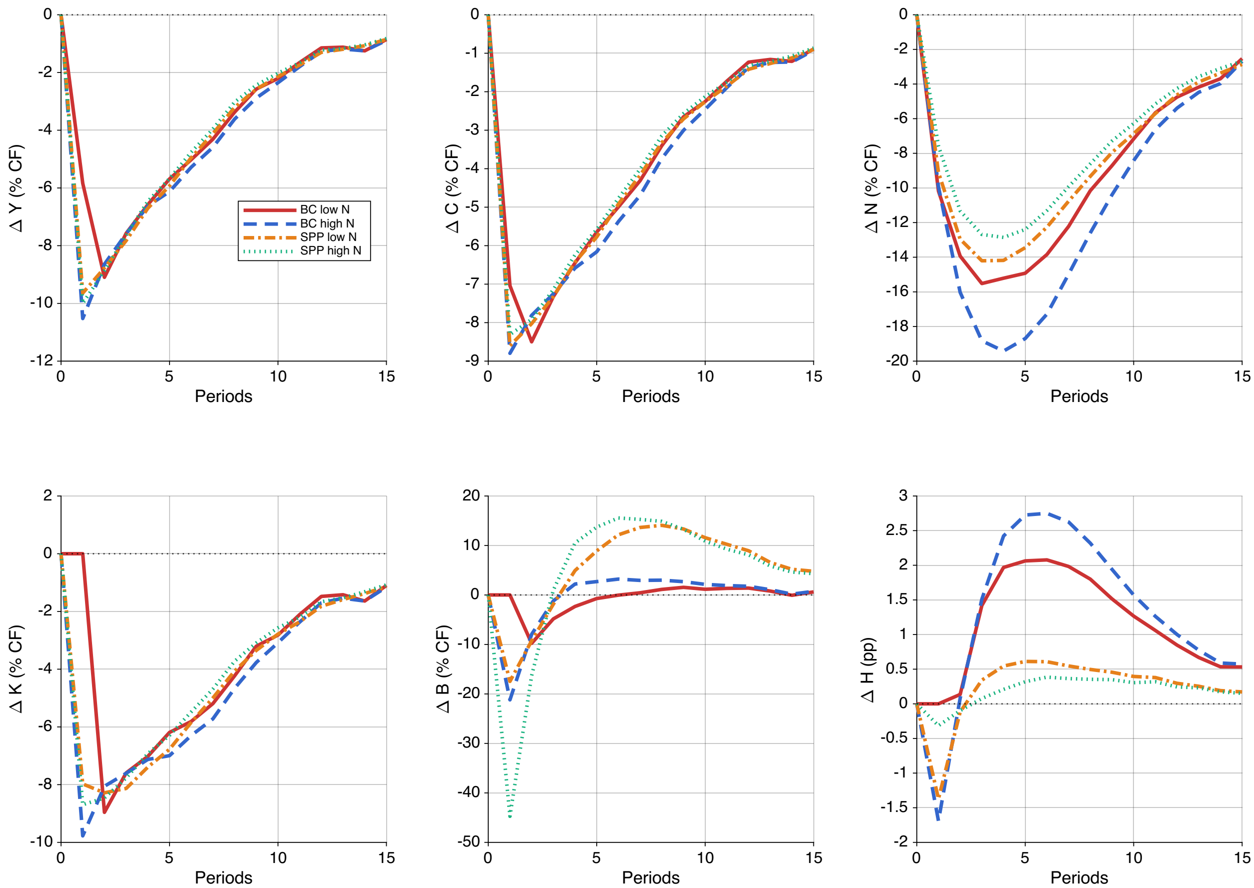

Figure 7: TFP shock GIRFs: CE vs. SPP.

Notes: CE low N_0 (red solid), CE high N_0 (blue dashed), SPP low N_0 (orange dash-dot), SPP high N_0 (teal dotted). Both CE and SPP are conditioned on the CE's 10th and 90th percentiles of N. GIRF expressed as percentage difference relative to the counterfactual (no-shock) path.

in both directions). This is the prices-versus-quantities trade-off of Weitzman (1974) applied to macroprudential policy: the price instrument is welfare-superior for the fragility margin but sacrifices the incidental TFP stabilization that the binding quantity constraint provided. Section 5 discusses the mechanisms further.

4.5 Ergodic distributions and welfare

The ergodic distributions are sharply separated: the CE occupies $N \in [0.009, 0.042]$, the SPP occupies $N \in [0.021, 0.072]$. The two economies effectively inhabit different regions of the state space. The planner's recession floor ($N \approx 0.021$) exceeds the CE's typical recession equity, keeping the economy above the leverage kink. Both economies achieve similar mean consumption; the welfare gain comes not from higher average output but from eliminating the left tail—the rare but costly episodes in which the leverage constraint

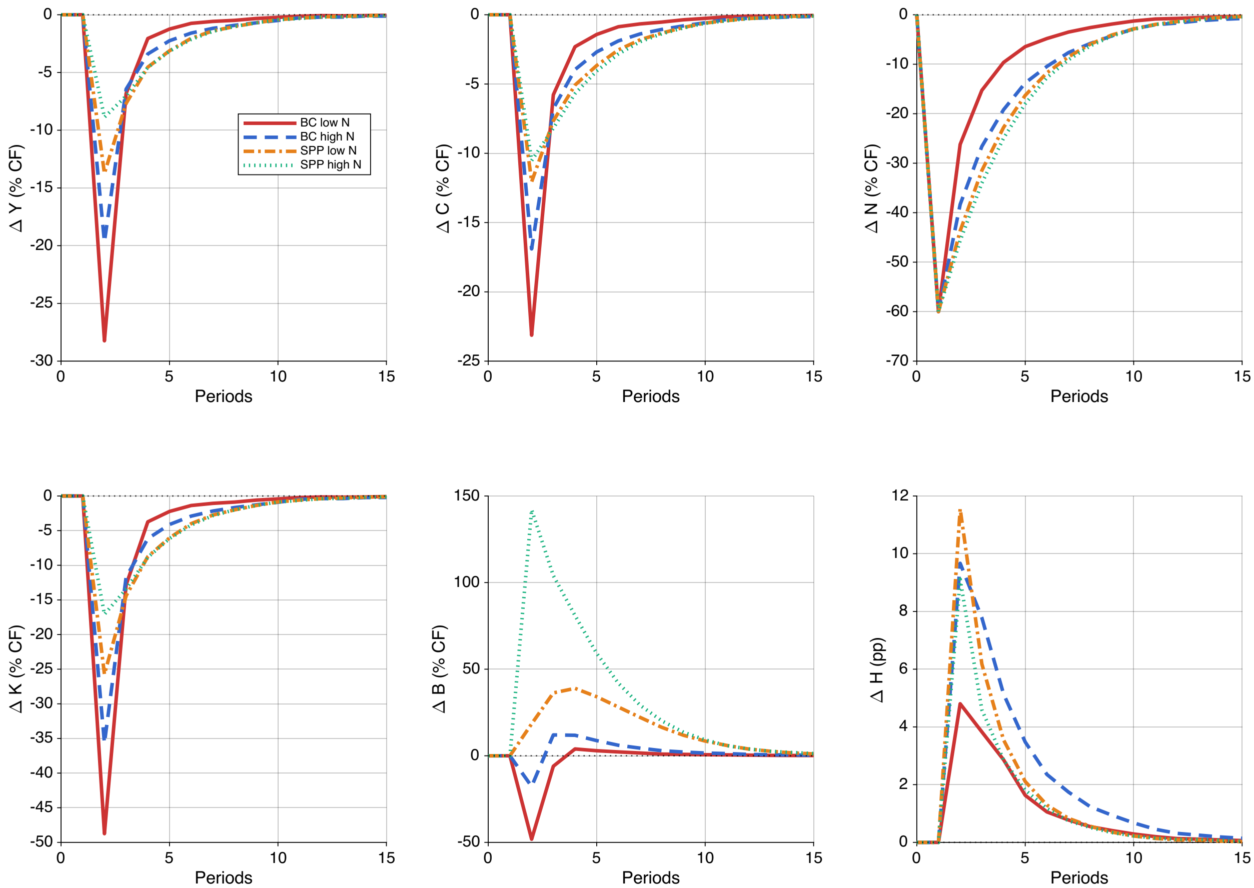

Figure 8: Hazard shock GIRFs: CE vs. SPP (expansion).

Notes: CE low N_0 (red solid), CE high N_0 (blue dashed), SPP low N_0 (orange dash-dot), SPP high N_0 (teal dotted). GIRF expressed as percentage difference relative to the counterfactual (no-shock) path.

binds and deleveraging spirals amplify distress.

Table 4 reports business cycle moments. The planner maintains higher mean equity, lower mean hazard ($\bar{\mathcal{H}}$), and lower crisis frequency (2.1% vs. 8.9%), while mean output and consumption are similar. A telling difference is in the higher moments: the CE displays strongly negative skewness in K, N, and H (reflecting occasional sharp contractions when the leverage constraint binds), whereas the SPP substantially compresses these left tails. The planner trades a small reduction in average dividends for a large reduction in downside risk—a classic insurance motive. The consumption-equivalent variation (CEV) is the permanent proportional increase in composite consumption $\widetilde{C}_t = C_t - \eta(H_t^S)^{1+1/\chi}/(1+1/\chi)$ that makes the CE household indifferent to the SPP allocation:

$$\lambda_{\text{CEV}} = \exp(\bar{u}_{\text{SPP}} - \bar{u}_{\text{CE}}) - 1 \tag{18}$$

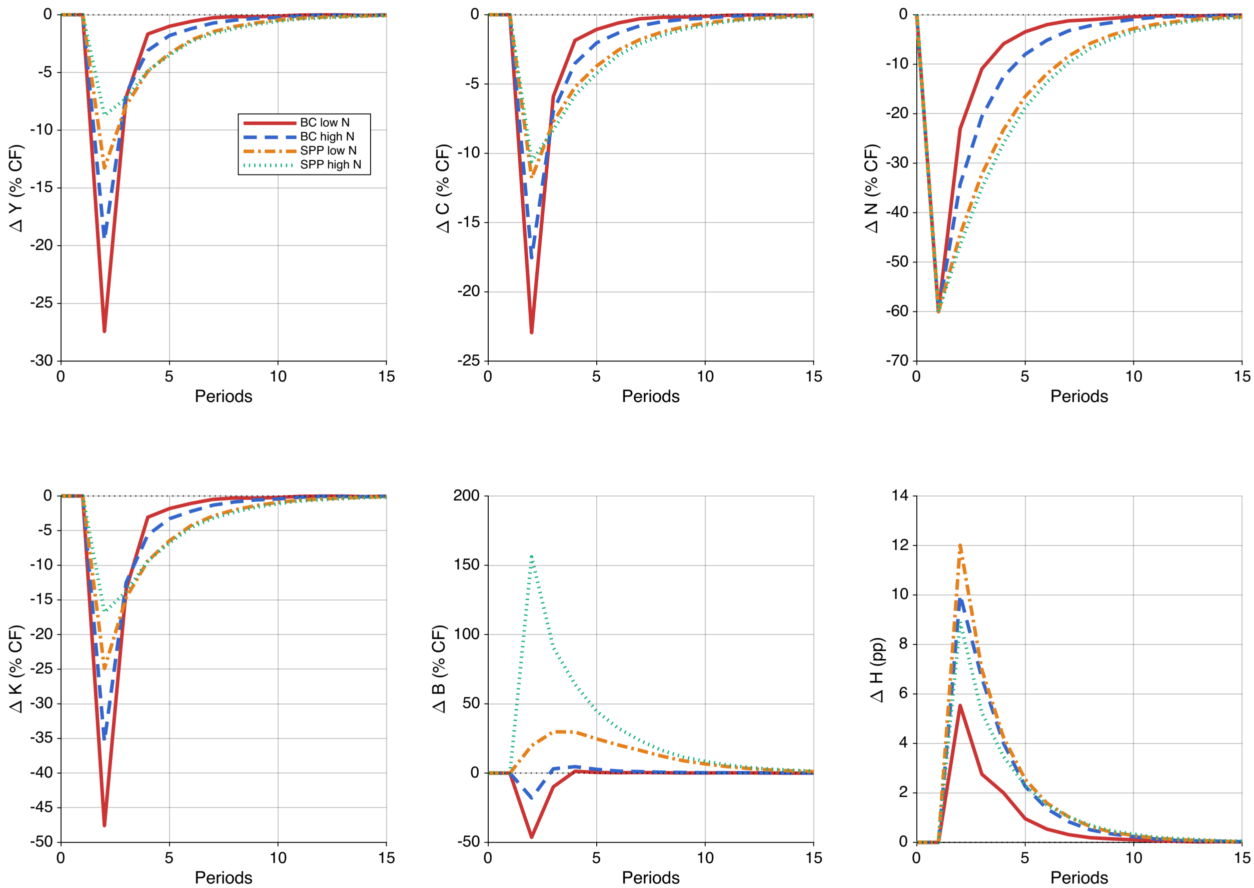

Figure 9: Hazard shock GIRFs: CE vs. SPP (recession).

Notes: Same construction as Figure 8; both paths start in recession.

where $\bar{u} = T^{-1} \sum_t \log \widetilde{C}_t$ is the average per-period utility over the ergodic path.

Table 4: Business Cycle Moments: CE vs. SPP

	Competitive Equilibrium			Social Planner		
Variable	Mean	Std. Dev.	Skew.	Mean	Std. Dev.	Skew.
Y	0.3255	0.0311	-1.362	0.3290	0.0222	-0.333
C	0.2216	0.0190	-1.205	0.2239	0.0145	-0.328
K	0.1012	0.0146	-2.124	0.1030	0.0087	-1.004
H	0.5785	0.0288	-1.532	0.5820	0.0198	-0.407
N	0.0334	0.0085	-0.909	0.0559	0.0125	-1.144
B	0.0452	0.0059	-1.492	0.0241	0.0069	$+1.104$
N/L	0.4184	0.0611	-0.131	0.6909	0.1160	-1.545
R^K	1.0743	0.1008	$+3.103$	1.0563	0.0380	$+4.402$
R^B	1.0452	0.0059	-1.492	1.0241	0.0069	$+1.104$
$\mathcal{H}$	0.0888	0.0344	$+0.772$	0.0208	0.0282	$+3.156$
Crisis frequency (%)		8.88			2.08	
CEV (%)		—			$+1.0819$	

Notes: Statistics computed over the ergodic distribution ($T = 5001$, burn-in $= 500$). CEV: consumption-equivalent variation of moving from CE to SPP.

5 Discussion

5.1 The capitalization dilemma

The saddle path decomposition in Section 4.3 maps the two individually well-known channels into a single geometric framework, making their interaction transparent.

The *credit exposure channel*—well documented in the financial accelerator literature—operates through the across-saddle gap $\Delta C(N)$. Higher bank equity supports more intermediation, making the economy more productive but also more exposed to TFP fluctuations. This is a "sowing the seeds" mechanism: credit booms during expansions simultaneously raise output and build vulnerability to productivity reversals. Since $\Delta C(N)$ widens with N, longer booms generate mechanically deeper busts. This channel is symmetric across the CE and SPP because both economies face the same intermediation technology.

The *fragility buffer channel*—studied in the banking crisis literature—operates through along-saddle displacement and the leverage kink. Lower initial equity means a distress event pushes the economy into the steep, constrained region of the saddle path, where deleveraging amplifies the initial equity loss. This channel is asymmetric across regimes because it depends on the *distance* from the kink, which the planner—but not the decentralized bank—can control. The market failure is a compositional externality: each bank's leverage choice is privately optimal, but the resulting aggregate capital ratio determines the system-wide failure probability that all banks face.

The unified framework reveals a tension that is invisible when the channels are studied separately: bank equity is the single state variable through which both channels operate, and any policy that adjusts it necessarily affects both. This has a direct implication for capital regulation. Tightening leverage requirements (raising ϕ_1) reduces credit exposure by compressing balance sheets and lowering $\Delta C(N)$, but also limits equity accumulation during expansions, potentially leaving the economy more exposed to distress. The regulator faces a frontier, not a free lunch: moving the economy rightward along the saddle path (more equity, less fragility) necessarily steepens the across-saddle gap (more TFP exposure).

In policy terms, the outcome is non-monotonic: extra capital makes rare crisis losses much less likely and less severe, but it can also make the economy more cyclically sensitive to ordinary real shocks, so that the marginal benefit of additional capital eventually comes with a visible stabilization cost.

The nature of this trade-off, however, depends critically on the *instrument* used to correct the fragility externality. The SPP internalizes the externality through a state-contingent Pigouvian tax $\tau(X)$ that raises the marginal cost of deposits. This tax *substitutes* for the leverage constraint: at the planner's allocation, the constraint shadow value λ falls sharply (from 0.066 in the CE to 0.003 in the SPP at the same equity level N), because the tax has already induced the bank to reduce leverage voluntarily. The bank no longer needs the hard cap to restrain its deposit-taking—the price signal does the work.

This substitution creates a subtle but genuine version of the capitalization dilemma. In the CE, the binding leverage constraint acts as a *straitjacket*: it prevents the bank from expanding in good times, but it also limits how far the bank can contract in bad times. The constraint inadvertently stabilizes the economy's response to TFP fluctuations by compressing the range of intermediation volumes. The planner's Pigouvian tax removes this rigidity. With the constraint nearly slack, the bank is free to adjust its balance sheet in response to TFP—expanding more aggressively in expansions, but also contracting more sharply in recessions. The result is a wider across-saddle gap for the SPP than for the CE at the same equity level: the proportional output decline from a negative TFP shock is larger under the planner's allocation, not because the planner intermediates more at a given N, but because the bank's *flexibility to respond* is greater when the hard constraint is replaced by a smooth price.

This mechanism connects to a classic insight in the regulation literature. Weitzman (1974) showed that price and quantity instruments yield different welfare outcomes when the relevant functions are nonlinear. Here, the nonlinearity is in the hazard function (6): the marginal cost of leverage rises steeply as the capital ratio falls. The Pigouvian tax—a price instrument—tracks this nonlinearity precisely, providing the optimal correction at every state. The leverage constraint—a quantity instrument—provides a cruder but more rigid correction that happens to stabilize TFP responses as a by-product. The capitalization dilemma is the cost of moving from the rigid instrument to the precise one: the planner gains fragility protection but loses the incidental TFP stabilization that the binding constraint provided.

The dilemma has an important temporal dimension. Ex ante, the planner's allocation is welfare-maximizing: no alternative policy can deliver higher expected lifetime utility. But ex post—conditional on the realized sequence of shocks—the competitive equilibrium can outperform the planner along particular sample paths. During prolonged expansions without bank failures, the CE economy produces more output and higher consumption because it is not holding the extra equity buffer that the planner demands. The planner's higher capital requirement acts as an insurance premium: it depresses dividends and credit intermediation in normal times, and its payoff materializes only when crises actually occur. In a fortunate history where no distress events strike, the premium is paid but never collected. This ex-post reversal is not a deficiency of the planner's policy but a defining feature of insurance: any policy that reduces tail risk must sacrifice expected payoffs in the states where the insured event does not occur. For the central banker, this creates a political economy challenge—the costs of higher capital requirements are visible in every period, while the benefits are concentrated in rare crises that may not materialize during a policymaker's tenure.

5.2 Precautionary savings as macroprudential policy

The planner's first-order condition differs from the bank's by the Pigouvian wedge $\tau(X)$ in (15), which reflects the marginal effect of leverage on failure probability. This wedge generates three reinforcing consequences. First, SPP conditional steady states exceed CE values by 76% in recession (0.0537 vs. 0.0305) and 70% in expansion (0.0721 vs. 0.0424), reflecting substantially higher equity targets. Second, these higher equity buffers prevent distress events from triggering the leverage constraint; deposits remain stable (ΔB GIRF ≈ 0), averting the credit contraction that drives CE output losses. Third, the SPP's post-distress equity floor ($N \approx 0.021$) exceeds the CE floor ($N \approx 0.009$) by a factor of

2.3, placing the economy above the leverage kink even after distress and enabling recovery along the flat portion of the saddle path without the amplification that prolongs CE recessions.

These mechanisms are mutually reinforcing: higher equity targets enable deposit stabilization, which prevents the constraint from binding and ensures fast recovery. In the CE, the externality operates through a tragedy-of-the-commons logic: each bank rationally minimizes its own equity to maximize dividends, but the collective outcome is a fragile system in which every bank is exposed to crises that none would individually choose.

The calibrated model pins down the implied Pigouvian tax. The wedge $\tau(X)$ enters the deposit margin in the same units as a net interest rate: it represents the per-unit tax on deposits that would induce banks to replicate the planner's equity choice.

Table 5: Implied Pigouvian Tax

	τ
Mean	0.0394
$A = A_L$ (Recession)	0.0412
$A = A_H$ (Expansion)	0.0378
Std. Dev.	0.0277

Notes: The Pigouvian wedge $\tau(X)$ from (15), in net-rate units (same as $R^B - 1$). Statistics over the ergodic distribution.

The wedge is countercyclical, rising in recessions—when the capital ratio is low and the hazard function is steep—and falling in expansions. The key insight is what this tax *buys* in terms of the saddle-path geometry. The CE's recession CSS ($N = 0.0305$) sits right at the leverage kink, so a single distress event pushes the economy into the steep, constrained region where deleveraging spirals take hold. The planner's tax shifts both conditional steady states well above this zone: the SPP recession CSS ($N = 0.0537$) exceeds even the CE *expansion* CSS ($N = 0.0424$). The planner's worst-case anchor point already lies beyond the best-case anchor of the unregulated economy.

This separation from the kink is not a knife-edge property of the calibration but arises from a self-reinforcing mechanism in the structure of τ itself. The Pigouvian wedge is *endogenously amplifying*: it rises sharply as the economy approaches the constrained region, because proximity to the kink simultaneously steepens the hazard function $\partial\mathcal{H}/\partial B$ and widens the value gap $V(n') - V((1-\varphi)n')$ in (15). In the calibrated model, τ quadruples from its ergodic mean of 0.04 to roughly 0.15 when the leverage constraint binds. This sharp increase in the cost of deposits near the kink acts as an endogenous "guardrail": the closer the economy drifts toward the constrained region, the more aggressively the planner's wedge pushes back by penalizing leverage, creating a restoring force that is absent in the CE.

That said, the guardrail does not provide absolute protection. Sufficiently large or repeated distress events can still drive the SPP economy into the constrained region—the constraint binds approximately 2% of the time even under the planner's policy, compared with 20% in the CE. But when it does bind, the starting point matters enormously. In the CE, the economy enters the constrained zone from a low equity base ($N \approx 0.009$), deep in the steep portion of the saddle path where each unit of equity loss triggers further deleveraging. The planner's economy enters the same zone from a much higher base ($N \approx 0.021$), close to the kink's boundary rather than deep within it, enabling a faster exit. The constraint binds *briefly and mildly* rather than *persistently and severely*: the planner converts what are prolonged credit crunches in the CE into transient episodes that resolve within a few periods.

The magnitude of τ reflects this geometric logic. The wedge is large—comparable in size to the CE deposit spread $R^B - 1$—because the externality it corrects is severe: in the CE, the leverage constraint binds frequently, amplifying every distress event into a prolonged contraction. The standard deviation of τ (Table 5) is large relative to its mean, reflecting the highly nonlinear nature of the hazard function: the marginal value of equity as a crisis buffer rises sharply as the capital ratio declines toward the steep region of the sigmoid (6).

In a decentralized implementation, a regulator could replicate the planner's allocation by levying a state-contingent tax $\tau(X)$ on deposit issuance and rebating the proceeds lump-sum to households. The tax raises the effective cost of deposits from $R^B - 1$ to $R^B - 1 + \tau$, inducing banks to substitute toward equity financing and thereby internalizing the fragility externality. Because the wedge varies with the aggregate state $X = (N, A)$, a fixed capital surcharge would only approximate the planner's optimum; the time variation in τ provides a quantitative benchmark for the state-contingency that countercyclical capital buffers should exhibit.

For TFP shocks, however, the planner faces the same technology constraint. Productivity shocks do not trigger failures, so the fragility externality is not activated and the planner cannot improve upon the market outcome. This clarifies the scope of macroprudential policy: it is effective where the market failure lies—in the systemic undervaluation of equity as a buffer against financial distress—and offers no advantage where shocks originate in the real sector.

5.3 Leverage limits vs. Pigouvian taxation

The Pigouvian tax $\tau(X)$ is a state-contingent instrument that is informationally demanding: it requires the regulator to observe the aggregate state and compute the marginal externality in real time. A simpler alternative is to tighten the leverage constraint directly by lowering ϕ_1. This exercise isolates what a blunt, time-invariant capital requirement can and cannot achieve.

Figure 10 compares the conditional saddle paths under three regimes: the baseline CE ($\phi_1 = 1.5$), a tighter leverage constraint ($\phi_1 = 0.8$), and the SPP. Tightening the constraint shifts both saddle paths rightward, raising the conditional steady states and moving the economy away from the kink—qualitatively similar to the Pigouvian tax. However, the mechanism differs in two important respects.

First, the tight constraint operates by *compressing credit across all states*. By capping deposits at a lower multiple of equity, it forces banks to shrink their balance sheets even when the economy is well-capitalized and the fragility externality is negligible. This flattens the saddle paths, reducing the across-saddle gap $\Delta C(N)$ and dampening output in normal times. The SPP, by contrast, achieves its equity targets through a state-contingent wedge that is small when the economy is healthy and large only near the kink. The planner's saddle paths retain their slope—credit intermediation is unimpaired in good states—while selectively steepening the penalty for leverage when it matters.

Second, the tight constraint cannot fully replicate the planner's allocation because it addresses the *symptom* (excessive leverage) rather than the *cause* (the uninternalized hazard externality). The constraint binds 13% of the time under the tight policy, versus 2% under the SPP. When it binds, it prevents deleveraging spirals—but it also prevents the economy from reaching the efficient level of intermediation. The SPP achieves lower binding frequency with higher mean output, revealing the efficiency cost of the blunt instrument.

The comparison highlights a general principle: leverage limits and Pigouvian taxes are complementary rather than substitutable instruments. Leverage limits provide a robust floor that does not require real-time state observation, but at the cost of compressing credit intermediation uniformly. The Pigouvian wedge targets the externality precisely but requires the regulator to track the aggregate state. In practice, a well-designed macroprudential framework would combine a time-invariant leverage floor (a relaxed version of the tight constraint) with a countercyclical buffer (approximating the state-contingent wedge) that activates as the system approaches the fragile region. The remainder of this section formalizes both ideas: Section 5.4 traces the policy frontier as ϕ_1 varies, and Section 5.5 derives a macroprudential rule from the planner's solution and interprets its

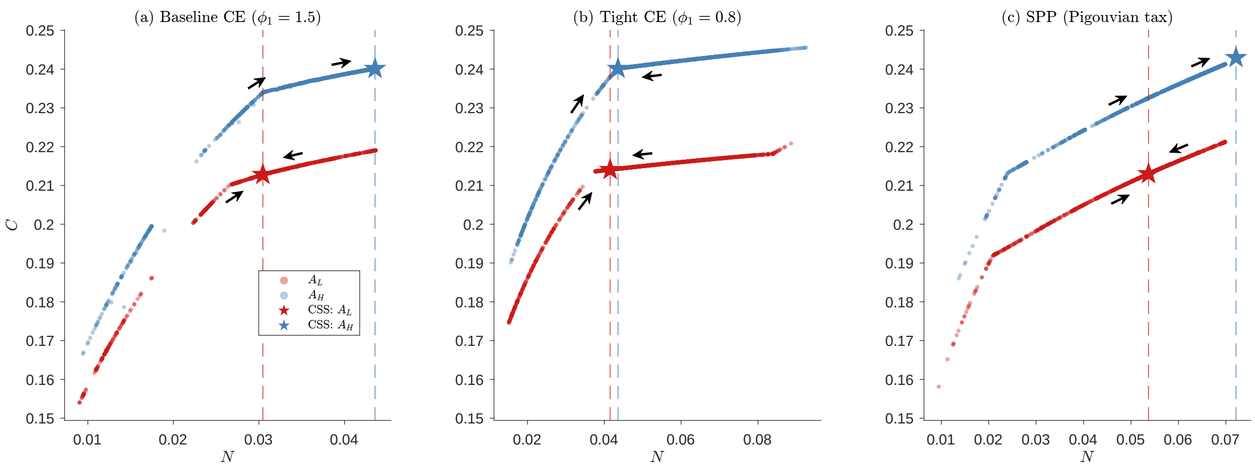

Figure 10: Conditional saddle paths: baseline CE, tight leverage constraint, and SPP.

Notes: Saddle paths under frozen TFP regimes. Stars mark conditional steady states. Panel (a): baseline CE ($\phi_1 = 1.5$). Panel (b): tight CE ($\phi_1 = 0.8$). Panel (c): SPP. All panels share common axes for direct comparison. The tight constraint shifts saddle paths rightward but also compresses them vertically, reflecting reduced credit intermediation.

structure.

5.4 Comparative statics of leverage policy

Before turning to state-dependent instruments, we use a fixed leverage limit as a policy-frontier experiment: it asks how far a simple, time-invariant rule can move the economy toward the planner's allocation and what efficiency cost it pays relative to the state-contingent Pigouvian benchmark. Table 6 reports steady-state and ergodic outcomes as ϕ_1 varies from 0.50 (tight) to 2.00 (loose), holding all other parameters at their baseline values. Three regularities emerge.

First, tightening the constraint monotonically reduces fragility. The mean hazard rate falls from roughly 10% at $\phi_1 = 2.00$ to below 3% at $\phi_1 = 0.50$, and crisis frequency declines proportionally. The mechanism is direct: a lower ϕ_1 caps deposits at a tighter multiple of equity, raising the *ergodic* capital ratio N/L—reflected here in the falling mean hazard $\bar{\mathcal{H}}$—and pushing the economy into the flat region of the sigmoid hazard function (6). (The first column of Table 6 reports the *deterministic* steady-state ratio, which is invariant to ϕ_1 by construction.)

Second, the binding frequency of the constraint is highly nonlinear in ϕ_1. At $\phi_1 = 0.50$ the constraint binds roughly two-thirds of the time, acting as a near-permanent cap on intermediation. At the baseline $\phi_1 = 1.50$ it binds 20% of the time—only during distress episodes. This nonlinearity implies that a regulator setting a constant leverage limit faces a steep trade-off: small reductions in ϕ_1 from the baseline dramatically increase the fraction of periods with binding credit rationing.

Third, welfare improves monotonically as ϕ_1 falls within the range considered (CEV reaches $+1.6\%$ at $\phi_1 = 0.50$), suggesting that the baseline leverage limit is too permissive from a welfare perspective. Indeed, a sufficiently tight constant cap attains a higher *ergodic* welfare gain than the Pigouvian SPP ($+1.6\%$ versus $+1.08\%$): the planner corrects the fragility externality at the baseline cap but does not re-optimize the cap *level*, so the SPP is the efficiency benchmark for the externality rather than the global optimum over leverage limits. The welfare gains from a tighter constant ϕ_1 nonetheless come from a fundamentally different source than the SPP's Pigouvian correction: the constant constraint compresses credit *uniformly* across all states, whereas the SPP targets the externality *selectively*. A constant ϕ_1 that binds two-thirds of the time reduces the hazard but also prevents the economy from reaching efficient intermediation levels in states where fragility is low. This distinction motivates the search for a state-dependent instrument.

Table 6: Comparative Statics: Leverage Slope ϕ_1

ϕ_1	N/L (det. SS)	$\bar{\mathcal{H}}$	Binding (%)	$\sigma(Y)$	$\bar{Y}$	CEV (%)
0.50	0.4975	0.0277	66.8	0.0258	0.3310	$+1.5559$
0.75	0.4975	0.0437	27.3	0.0281	0.3286	$+0.8478$
1.00	0.4975	0.0582	20.3	0.0294	0.3274	$+0.5096$
1.25	0.4975	0.0732	19.8	0.0304	0.3265	$+0.2532$
$1.50^{\dagger}$	0.4975	0.0887	17.8	0.0311	0.3255	$+0.0024$
1.75	0.4975	0.1033	18.1	0.0314	0.3246	-0.2417
2.00	0.4975	0.1190	20.0	0.0316	0.3236	-0.4844

Notes: †Baseline calibration. N/L (det. SS) is the deterministic steady-state capital ratio, invariant to ϕ_1 by construction; the *ergodic* capital ratio rises as ϕ_1 tightens, reflected in the falling $\bar{\mathcal{H}}$ (mean ergodic hazard). Binding: fraction of periods with an active leverage constraint. CEV: consumption-equivalent welfare change relative to baseline.

Unlike the optimal state-contingent rule, a constant tighter leverage ratio achieves some stability at the cost of permanently stunted credit and still does not prevent the constraint from binding as effectively: even the tight fixed rule binds about 13% of the time, compared with roughly 2% under the SPP. This supports the practical case for adding a countercyclical component to static capital requirements, so that policy can tighten near fragile states without unnecessarily repressing credit when the system is safely capitalized.

5.5 A macroprudential rule from the planner's solution

We now turn from the policy-frontier experiment to implementation. The question is not whether the fixed rule improves on laissez-faire, but how closely a practical, state-contingent leverage rule can approximate the planner's first-best Pigouvian correction. We present the rule in two layers. First, we give the intuition: the rule separates the financial cycle from the business cycle, tightening when balance-sheet raises fragility and loosening when strong fundamentals reduce the marginal externality. Second, we show how the rule is constructed from the planner's allocation by translating the Pigouvian wedge into an SPP-implied leverage tightness and estimating a parsimonious regression. Aggregate credit-gap rules such as the Basel III CCyB (Basel Committee on Banking Supervision, 2010) collapse these two channels into a single credit-growth response and hide the structure. The closest antecedent is the optimized macroprudential Taylor rule of Bianchi and Mendoza (2018), which targets a Pigouvian tax on debt in a small open economy. Our rule differs on two dimensions. The instrument is a leverage coefficient on bank equity rather than a debt tax, mapping directly to Basel-style capital requirements. The hazard function is calibrated empirically using the Jordà et al. (2017) crisis database. Separately from these differences, the global solution also matters: it estimates the rule over the nonlinear region around the leverage kink, where local approximations would understate the state dependence.

5.5.1 Intuition and implementation

The implementable rule has a simple policy interpretation. It asks regulators to adjust the leverage coefficient using two separate signals rather than a single composite credit-gap indicator. The financial-cycle signal tightens the rule when aggregate bank equity and balance-sheet capacity are high, because this is when credit expansion can erode the sector-wide capital ratio. The business-cycle signal loosens the rule when TFP is high, because strong fundamentals reduce the marginal fragility externality and make additional intermediation less dangerous.

In practice, this means the rule distinguishes two cases that a credit-to-GDP trigger can conflate. Credit growth driven by expanding bank balance sheets calls for tighter leverage limits; credit growth driven by stronger fundamentals can justify looser limits. The technical construction below shows how we extract this rule from the planner's allocation.

5.5.2 Deriving the rule from the planner's allocation

From the SPP solution, define the state-contingent leverage tightness:

$$\phi_1^*(N, A) \;:=\; \frac{B^{\text{SPP}}(N, A) - \phi_0}{N} \tag{19}$$

This quantity answers the following question: if the CE bank faced a leverage constraint with slope ϕ_1^* at state (N, A), at what level would the constraint bind to exactly match the SPP's deposit choice?

The ergodic distribution of ϕ_1^* reveals the gap between the CE and SPP in terms of the leverage instrument. At the SPP's ergodic distribution, ϕ_1^* averages 0.32—roughly one-fifth of the baseline CE value of 1.50. The standard deviation is 0.31, reflecting substantial state dependence. ϕ_1^* is near zero in recessions and rises to the CE cap of 1.50 in high-TFP states. This does not mean that low equity tightens the rule. Holding TFP fixed, low equity in fact *loosens* it via $\gamma_N < 0$. The tight recession values arise because the low-TFP channel (via $\gamma_A > 0$) dominates the offsetting low-equity channel. Interpreting γ_N therefore requires conditioning on TFP: it is a *partial* effect. Crucially, $\phi_1^* > 0$ everywhere—the planner never requires $B < \phi_0$—so the one-sided nature of the constraint is not a binding limitation.

5.5.3 Estimated rule and interpretation

The SPP-implied tightness $\phi_1^*(N, A)$ varies smoothly with the aggregate state, suggesting that a simple rule can approximate it well. We fit a linear macroprudential rule by OLS:

$$\phi_1^*(N, A) \;\approx\; \bar{\phi}_1 + \gamma_N \left(\frac{N}{N_{\text{ss}}} - 1 \right) + \gamma_A (A - 1) \tag{20}$$

where the regressors are the fractional deviation of equity from its steady state and the TFP deviation. The estimation uses observations from the SPP's post-burn-in ergodic simulation ($T = 5{,}001$ periods).

Table 7 reports the results. The OLS fit achieves an R^2 of 0.927; adding an interaction term raises it only modestly to 0.931, confirming that the linear specification captures the essential structure. The high explanatory power is itself a finding: the planner's complex, forward-looking policy—which solves a dynamic programming problem over the full state space—can be well approximated by a regulator who observes only two aggregate indicators and adjusts the leverage limit according to a fixed linear formula.

Table 7: SPP-Implied Macroprudential Rule

	Estimate
$\bar{\phi}_1$ (intercept)	0.687
γ_N (financial cycle: equity response)	-1.055
γ_A (business cycle: TFP response)	$+5.358$
R^2	0.927
R^2 (with interaction)	0.931

Notes: OLS regression of the SPP-implied leverage tightness $\phi_1^*(N, A) := (B^{\text{SPP}} - \phi_0)/N$ on $(N/N_{\text{ss}} - 1)$ and $(A - 1)$, using the SPP's ergodic simulation ($T = 5{,}001$). The interaction specification adds $(N/N_{\text{ss}} - 1)(A - 1)$ as a regressor. The negative γ_N tightens leverage when credit expands; the positive γ_A loosens it when fundamentals are strong.

5.5.4 Interpreting the estimated coefficients

The estimated coefficients reveal a fundamental decomposition of the planner's prudential stance into two orthogonal components:

- **Financial cycle** ($\gamma_N = -1.06$): The rule *tightens* the constraint when bank equity rises above its steady-state level and *loosens* it when equity falls below—a "lean against the credit cycle" motive that imposes a *speed limit* on aggregate balance-sheet expansion. The sign may appear counter-intuitive, since well-capitalized banks are individually less fragile; we therefore read γ_N as a descriptive summary of the SPP's implied leverage tightness rather than a single structural channel. What is unambiguous is that the externality is what makes intervention warranted: each atomistic bank takes the aggregate failure hazard as given, ignoring its contribution to the sector-wide capital ratio that determines it.

- **Business cycle** ($\gamma_A = 5.36$): The rule *loosens* the constraint in expansions ($A > 1$) and tightens it in recessions ($A < 1$). This sign appears pro-cyclical, seemingly at odds with counter-cyclical capital buffer (CCyB) logic. The resolution lies in the structure of the Pigouvian tax (15). The tax $\tau \propto \frac{\partial \mathcal{H}}{\partial B} \cdot \mathbb{E}[V(n') - V((1-\varphi)n')]$ depends on two factors: the sensitivity of the hazard to leverage ($\partial \mathcal{H}/\partial B$) and the welfare cost of failure ($V(n') - V((1-\varphi)n')$). In expansions, both factors are small. Higher TFP raises intermediation profits, pushing the economy into the flat region of the hazard sigmoid where $\partial \mathcal{H}/\partial B$ is low; simultaneously, the value gap $V(n') - V((1-\varphi)n')$ narrows because the economy can recover quickly from distress when fundamentals are strong. The planner can therefore afford to loosen the constraint, allowing banks to intermediate more when doing so is least dangerous. In recessions, both factors rise sharply: the hazard function steepens as the economy approaches the kink, and the welfare cost of failure increases because recovery is slow when profits are compressed. The planner tightens accordingly.

The main implication is the *separation*: optimal macroprudential policy should track the financial cycle and the business cycle independently, because the two cycles call for opposite responses. A regulator who observes rising output and rising credit simultaneously receives conflicting signals. Output growth says "loosen—fundamentals are strong and the hazard is low"; credit growth says "tighten—balance sheets are expanding and systemic leverage is building." The rule (20) resolves this conflict quantitatively: each coefficient responds to its respective state variable with its own sign and magnitude.

This separation has direct implications for the design of the Basel III counter-cyclical capital buffer (CCyB). The CCyB responds to the credit-to-GDP gap. This gap is a single composite indicator that conflates the financial and business cycles.

Consider a recession with sticky credit. Credit-to-GDP rises mechanically as GDP falls, and the CCyB tightens. Our rule's TFP margin ($\gamma_A > 0$, $A < 1$) also tightens, because the marginal fragility externality is elevated when fundamentals are weak. But whether the *net* response is a tightening depends on the equity state. With $\gamma_N < 0$, the equity margin loosens when N falls below steady state. The composite CCyB cannot tell which channel is driving the signal.

Consider now a high-TFP expansion with ordinary balance-sheet growth. The CCyB triggers a tightening on credit-to-GDP. Our rule's TFP term pushes toward loosening, because strong fundamentals reduce the hazard externality. The equity term pushes toward tightening. The net response depends on which channel dominates—a distinction the composite cannot make.

By conditioning on N and A separately, the rule (20) lets the regulator distinguish credit growth driven by balance-sheet expansion (tighten via γ_N) from credit growth driven by strong fundamentals (loosen via γ_A).

5.5.5 The intercept: the planner's baseline stance

The intercept $\bar{\phi}_1 = 0.69$ deserves separate discussion. It is less than half the baseline CE value of 1.50, implying that the planner's *unconditional* leverage limit is substantially tighter than in the decentralized equilibrium. In the baseline calibration, $\phi_1 = 1.50$ is chosen so that the constraint binds approximately 20% of the time—a target motivated by the observation that financial constraints are not permanently binding. The planner's implied intercept of 0.69 would, if applied as a constant rule, cause the constraint to bind far more frequently.

This gap reflects the wedge between private and social returns to leverage. In the CE, banks choose leverage to maximize private franchise value, ignoring its effect on the aggregate hazard rate. The planner internalizes this externality and demands a tighter baseline—not because individual banks are imprudent, but because their collectively rational choices produce a systemically fragile outcome. The magnitude of the gap ($0.69/1.50 = 46\%$) measures the severity of the externality at the calibrated parameters: the planner would roughly halve the permitted leverage ratio.

5.5.6 Limitations of the leverage instrument

While the linear rule achieves a high R^2, it is important to emphasize what it does *not* accomplish. The leverage constraint controls deposits—one of two margins that the Pigouvian tax corrects. The other is equity retention: the choice of n'. In the SPP, the tax enters the envelope condition as $J_1^{\text{SPP}} = R^B + \lambda(1 + \phi_1) + \tau$, raising the marginal value of equity and inducing the bank to retain more. A binding leverage constraint provides an indirect bonus to equity retention ($\lambda > 0$ raises $J_1 = R^B + \lambda(1 + \phi_1)$), but this bonus is determined by the constraint's shadow value, not by the externality itself. Consequently, the bank's equity choice under the leverage rule differs from the planner's, and the resulting allocation does not replicate the SPP.

This "one instrument, two margins" limitation implies that no leverage rule—however carefully calibrated—can achieve the same outcome as the Pigouvian tax. The leverage instrument controls the *quantity* of deposits but not the *price* of the externality. The linear rule (20) is therefore best understood not as a substitute for the Pigouvian tax but as a *characterization* of what the planner would want the leverage instrument to do: how it should move with bank equity, how it should respond to fundamentals, and where the baseline should be set. It provides a structural benchmark against which actual regulatory rules can be evaluated.

5.5.7 The value of making the rule common knowledge

Finally, we ask whether the rule's welfare content depends on banks *anticipating* its state-contingent structure, as opposed to merely facing its realized values. Comparing two rational-expectations equilibria under the same enforced rule—one in which banks correctly condition on $\phi_1(N, A)$, the other in which they treat ϕ_1 as an i.i.d. draw around its unconditional mean—we find that making the structure common knowledge is robustly welfare-improving but second-order: a consumption-equivalent gain of $+0.061\%$ (95% confidence interval $[+0.055\%, +0.068\%]$ across $n = 10$ simulation seeds). The gain operates through intermediation efficiency rather than financial stability: anticipating that the cap loosens in expansions, banks time their borrowing slightly better, while the mean failure probability is essentially unchanged across the two economies.

5.6 Anatomy of the SPP–CE capitalization gap

Proposition 1 establishes that the planner *always* accumulates more equity than the competitive equilibrium: $N^{\text{SPP}} > N^{\text{CE}}$. Because TFP shocks are exogenous and symmetric across agents, no individual bank's actions can alter them, so there is no externality to correct on that margin. The failure hazard, by contrast, is an endogenous aggregate outcome: each bank takes $\mathcal{H}$ as given, yet its leverage choice contributes to the sector-wide capital ratio that determines $\mathcal{H}$. The Pigouvian wedge $\tau > 0$ prices exactly this missing feedback. Consequently, the *direction* of optimal policy is unambiguous—more equity is always warranted—and the operative question is *how much*. This section decomposes the quantitative magnitude of the SPP–CE gap into three structural channels and shows, via a counterfactual exercise, that the gap is highly sensitive to two primitive parameters: the severity of bank failure (φ) and the volatility of TFP shocks (σ_A).

5.6.1 Three-channel decomposition

The envelope conditions for the CE and SPP are:

$$J_1^{\text{CE}}(N; X) = R^B + \lambda\,(1 + \phi_1), \tag{21}$$
$$J_1^{\text{SPP}}(N; X) = R^B + \lambda\,(1 + \phi_1) + \tau. \tag{22}$$

At each economy's respective ergodic distribution, the Euler equation pins down the average marginal value of equity. The gap $J_1^{\text{SPP}} - J_1^{\text{CE}}$ can be decomposed into three channels evaluated at the ergodic means—an *ergodic-accounting* decomposition across the two economies' *distinct* stationary distributions, not a same-state marginal comparison:

$$\underbrace{\mathbb{E}\left[J_1^{\text{SPP}}\right] - \mathbb{E}\left[J_1^{\text{CE}}\right]}_{\text{total gap}} = \underbrace{\Delta R^B}_{\text{deposit rate}} + \underbrace{\Delta[\lambda\,(1 + \phi_1)]}_{\text{shadow value}} + \underbrace{\mathbb{E}[\tau]}_{\text{Pigouvian wedge}}, \tag{23}$$

where $\Delta R^B \equiv \mathbb{E}[R^{B,\text{SPP}}] - \mathbb{E}[R^{B,\text{CE}}]$ and $\Delta[\lambda(1 + \phi_1)]$ is defined analogously. Although Proposition 1 guarantees $\tau > 0$, the other two channels are *negative* in equilibrium, partially offsetting the Pigouvian wedge:

- **[+] Pigouvian wedge** ($\tau > 0$): The planner internalizes that higher equity reduces the failure hazard, creating a positive marginal value absent in the CE. This channel pushes the SPP toward *more* equity.

- **[−] Deposit rate gap** ($\Delta R^B < 0$): The SPP restricts deposits to reduce leverage. With lower B, the deposit cost $\zeta'(B) = 2\zeta_2 B$ falls, so $R^B = 1 + \zeta'(B)$ is lower for the SPP. A lower deposit rate reduces the return on equity, pushing the SPP toward *less* equity.

- **[−] Shadow value gap** ($\Delta[\lambda(1 + \phi_1)] < 0$): The SPP's lower leverage means its constraint binds far less frequently (2% vs. 20% at baseline). The resulting lower average λ reduces the equity return, again pushing the SPP toward *less* equity.

Table 8 reports the decomposition at baseline calibration. The Pigouvian wedge ($\tau = 0.040$) is positive but *smaller* than the combined offset from the deposit rate and shadow value channels ($-0.021 - 0.103 = -0.124$). The net gap is negative: $\mathbb{E}[J_1^{\text{SPP}}] < \mathbb{E}[J_1^{\text{CE}}]$. This is *consistent* with Proposition 1: the SPP holds more equity, and at higher N the marginal return to equity is lower due to diminishing returns from intermediation. The planner accumulates equity precisely *until* the combined return—inclusive of the Pigouvian bonus—falls to the household's required rate. The negative net gap reflects the fact that the SPP's equilibrium equity is high enough to exhaust the excess returns that the CE leaves on the table.

Table 8: Decomposition of the SPP–CE Return Gap

	Baseline ($\sigma_A = \pm 2\%$, $\varphi = 0.60$)	Counterfactual ($\sigma_A = \pm 5\%$, $\varphi = 0.20$)	Direction
Pigouvian wedge $\mathbb{E}[\tau]$	$+0.040$	$+0.014$	[+] more N
Deposit rate gap ΔR^B	-0.021	-0.006	[−] less N
Shadow value gap $\Delta[\lambda(1 + \phi_1)]$	-0.103	-0.009	[−] less N
Net gap	-0.084	-0.001	
$N^{\text{SPP}}/N^{\text{CE}} - 1$	$+67\%$	$+14\%$	

Notes: Ergodic means from the baseline ($\sigma_A = \pm 2\%$, $\varphi = 0.60$) and counterfactual ($\sigma_A = \pm 5\%$, $\varphi = 0.20$) calibrations. The Pigouvian wedge τ scales approximately linearly with φ; the shadow value gap depends on TFP volatility through the frequency with which the leverage constraint binds.

5.6.2 Counterfactual: high TFP volatility, low failure severity

To assess how much the SPP–CE gap depends on primitives, we re-solve the model under a counterfactual calibration that widens TFP volatility ($\sigma_A = \pm 5\%$ vs. baseline $\pm 2\%$) while reducing the bank failure severity ($\varphi = 0.20$ vs. baseline 0.60). Both changes *weaken* the forces that offset the Pigouvian wedge: lower φ shrinks τ (bank failures are less costly, so the externality weakens), and although wider TFP shocks raise the raw incidence of large draws, the CE responds by accumulating a larger precautionary equity buffer, so its leverage constraint binds *less* often—compressing the shadow-value gap (as detailed below).

As the right column of Table 8 reports, the SPP–CE equity gap shrinks dramatically in the counterfactual: from 67% at baseline to 14%. The Pigouvian wedge falls from $\tau = 0.040$ to 0.014—close to the combined deposit rate and shadow value offsets—so the SPP's additional equity retention is modest.

The mechanism operates through the interaction of the three channels. In the counterfactual, the Pigouvian wedge τ falls to approximately one-third of its baseline value, since $\tau \propto \varphi$: for small φ, $V(n') - V((1-\varphi)n') \approx \varphi \cdot V'(n') \cdot n'$, so τ scales nearly linearly with the failure severity parameter. Meanwhile, the CE leverage constraint binds less frequently (8% vs. 18%) despite wider TFP, because the CE itself accumulates more equity as a buffer against larger shocks. The SPP constraint still binds 4% of the time (vs. 2% at baseline), so the shadow value gap between the two economies shrinks dramatically (from -0.103 to -0.009). Finally, the deposit rate gap narrows because the SPP, facing a smaller externality, restricts deposits less aggressively.

In the limit $\varphi \to 0$, bank failures become costless, $\tau \to 0$, and the SPP collapses to the CE. The Pigouvian wedge is the single structural force separating the two economies, and its magnitude is governed by the severity of bank failure.

Because the Pigouvian wedge is strictly positive, the *direction* of optimal policy is never in doubt: the planner always wants more equity than the market provides. The economically important question is *how much more*. The three-channel decomposition (23) provides a structural answer. The case for large macroprudential capital buffers is strongest when bank failures are costly and TFP volatility is moderate—the configuration in which the Pigouvian wedge is largest relative to the offsetting deposit-rate and shadow-value channels. When failure costs are low (e.g., due to effective resolution regimes or deposit insurance), the Pigouvian wedge shrinks and the two offsetting channels nearly neutralize it, so the optimal buffer narrows from 67% to 14% of CE equity. The decomposition thus serves as a diagnostic for calibrating countercyclical capital buffers: the buffer should be large enough to close the Pigouvian gap but not so large as to overshoot the externality it corrects.

6 Conclusion

This paper develops a general equilibrium model that nests the credit exposure and fragility buffer channels—individually well understood in the literature—within a single framework. Doing so reveals a capitalization dilemma: bank equity amplifies TFP shocks through the credit exposure channel but mitigates financial shocks through the fragility buffer channel. Because both channels operate through the same state variable, a social planner who internalizes the fragility externality halves the damage from financial shocks while gaining no advantage for productivity shocks. This asymmetry delineates the scope of macroprudential policy: it is most valuable for financial fragility rather than real-sector fluctuations.

Several simplifying assumptions merit discussion. First, the representative bank abstracts from interbank markets and balance-sheet heterogeneity, which matter for contagion and the cross-sectional distribution of fragility. Second, the two-state TFP process suffices for the dual-role mechanism but limits the model's ability to match business cycle moments. Third, full depreciation ($\delta = 1$) eliminates physical capital dynamics; relaxing it would add capital as an endogenous state and enrich the interaction between bank equity and real investment. Fourth, the reduced-form hazard (6) does not microfound the failure mechanism; a strategic complementarity structure à la Goldstein and Pauzner (2005) would endogenize the hazard at the cost of additional complexity.

These limitations suggest natural extensions. Heterogeneous banks would allow study of how cross-sectional variation in capitalization affects systemic risk and bank-specific capital requirements. A richer shock process (e.g., persistent TFP, time-varying financial shock intensity) would improve quantitative realism. Combining the model with nominal rigidities would permit study of macroprudential–monetary policy interactions—in particular, whether the dual role extends to environments where the central bank faces a trade-off between output stabilization and financial stability.

From a policy perspective, we derive a linear macroprudential rule from the planner's solution: $\phi_1(N, A) = \bar{\phi}_1 + \gamma_N(N/N_{ss} - 1) + \gamma_A(A - 1)$, which approximates the SPP-implied leverage tightness with an R^2 of 0.93. The rule reveals a fundamental separation in the prudential stance. The financial cycle component ($\gamma_N < 0$) tightens leverage limits when aggregate bank equity rises, leaning against the credit cycle. The business cycle component ($\gamma_A > 0$) loosens the constraint in expansions, when strong fundamentals naturally push the economy away from the fragile region. These two responses carry opposite signs because the financial and business cycles move the fragility externality in opposite directions—a distinction that is lost in composite indicators such as the credit-to-GDP gap used in the Basel III CCyB framework. The rule provides a structural benchmark against which actual regulatory instruments can be evaluated, identifying where existing frameworks conflate distinct sources of risk and where the planner would prescribe differentiated responses.

The three-channel decomposition (Section 5.6) shows that the magnitude of the optimal buffer is sensitive to the severity of bank failure: when resolution regimes effectively limit failure costs, the Pigouvian wedge shrinks and the SPP–CE gap narrows from 67% to 14%, tempering the case for aggressive capital surcharges while preserving the qualitative direction of the policy prescription.

In summary, our model reveals a fundamental capitalization dilemma: higher bank capital reduces crisis frequency and severity dramatically, but can also make ordinary productivity-driven recessions slightly worse by increasing exposure to real shocks. The optimal policy therefore does not eliminate trade-offs; it operates on the frontier between resilience and macroeconomic volatility. The macroprudential rule we derive offers a practical way to move along that frontier, capturing most of the stability gains from higher capital while limiting the cost in normal times by tightening when balance-sheet expansion raises fragility and relaxing when strong fundamentals reduce the marginal externality. Ultimately, the paper provides a quantitative framework for understanding the trade-offs inherent in bank capital regulation and offers guidance on how macroprudential policy can safeguard financial stability without unduly constraining economic performance.

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